

Engineering Mathematics

Previous Year Questions • 1994–2026

LINEAR ALGEBRA

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Linear Algebra (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Linear Algebra

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7. Rank of a Matrix
8. Solution of Non-Homogeneous Equations
9. Solution of Homogeneous Equations
10. Eigen Values and Eigen Vectors
11. Properties of Eigen Values and Eigen Vectors
12. Vector Space, Subspace and Linear Span
13. Basis Vectors and Null Space

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 52 **MCQ:** 35 **MSQ:** 2 **NAT:** 15
1M: 31 **2M:** 21 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–3M(2025)

EE — Electrical Engineering

Questions: 55 **MCQ:** 42 **MSQ:** 3 **NAT:** 10
1M: 30 **2M:** 24 **Desc:** 1 **Avg Weightage** ~3M **Range:** 1M(2023)–8M(2026)

IN — Instrumentation Engineering

Questions: 36 **MCQ:** 27 **MSQ:** 2 **NAT:** 7
1M: 24 **2M:** 12 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2019)–3M(2023)

CS — Computer Science & IT

Questions: 57 **MCQ:** 34 **MSQ:** 6 **NAT:** 17
1M: 33 **2M:** 23 **Desc:** 1 **Avg Weightage** ~3M **Range:** 1M(2023)–4M(2025)

ME — Mechanical Engineering

Questions: 45 **MCQ:** 37 **MSQ:** 2 **NAT:** 6
1M: 30 **2M:** 15 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2022)

CE — Civil Engineering

Questions: 47 **MCQ:** 35 **MSQ:** 6 **NAT:** 6
1M: 31 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2019)–4M(2023)

CH — Chemical Engineering

Questions: 22 **MCQ:** 14 **MSQ:** 1 **NAT:** 7
1M: 13 **2M:** 8 **Desc:** 1 **Avg Weightage** ~2M **Range:** 1M(2026)–3M(2021)

PI — Production & Industrial Engineering

Questions: 25 **MCQ:** 22 **MSQ:** 0 **NAT:** 3
1M: 16 **2M:** 9 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q1.26.1 MSQ 2026 1M

Consider the matrix $M = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{bmatrix}$. Which of the following options is/ are TRUE if $\det(M) \neq 0$?

- **A.** $a = -\frac{1}{2}$ and $b = -\frac{1}{2}$
- **B.** $a = \frac{1}{2}$ and $b = \frac{1}{2}$
- **C.** $a = -3$ and $b = 0$
- **D.** $a = \frac{1}{2}$ and $b = -3$

Q1.25.1 MCQ 2025 1M

Consider the matrix A below:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \gamma \end{bmatrix}$$

For which of the following combinations of α, β, and γ, is the rank of A at least three?

(i) $\alpha = 0$ and $\beta = \gamma \neq 0$.

(ii) $\alpha = \beta = \gamma = 0$.

(iii) $\beta = \gamma = 0$ and $\alpha \neq 0$.

(iv) $\alpha = \beta = \gamma \neq 0$.

A. Only (i), (iii), and (iv)

B. Only (iv)

C. Only (ii)

D. Only (i) and (iii)

Q1.25.2 NAT 2025 2M

Consider the vectors

$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 3\sqrt{2} \end{bmatrix}.$$

For real-valued scalar variable x, the value of

$$\min_x \|\mathbf{a}x - \mathbf{b}\|_2$$

is _____ (rounded off to two decimal places).

$\|\cdot\|_2$ denotes the Euclidean norm, i.e., for $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$, $\|\mathbf{y}\|_2 = \sqrt{y_1^2 + y_2^2}$.

Q1.24.1 NAT 2024 1M

Let $\mathbb{R}$ and $\mathbb{R}^3$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors

$$\{[2 \quad -3 \quad \alpha], \quad [3 \quad -1 \quad 3], \quad [1 \quad -5 \quad 7]\}$$

does not form a basis of $\mathbb{R}^3$ is _____.

Q1.24.2 MSQ 2024 2M

Consider the matrix $\begin{bmatrix} 1 & k \\ 2 & 1 \end{bmatrix}$, where k is a positive real number. Which of the following vectors is/are eigenvector(s) of this matrix?

- **A.** $\begin{bmatrix} 1 \\ -\sqrt{2/k} \end{bmatrix}$
- **B.** $\begin{bmatrix} 1 \\ \sqrt{2/k} \end{bmatrix}$
- **C.** $\begin{bmatrix} \sqrt{2k} \\ 1 \end{bmatrix}$
- **D.** $\begin{bmatrix} \sqrt{2k} \\ -1 \end{bmatrix}$

Q1.23.1 MCQ 2023 1M

Let the sets of eigenvalues and eigenvectors of a matrix B be $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{v_k \mid 1 \leq k \leq n\}$, respectively. For any invertible matrix P, the sets of eigenvalues and eigenvectors of the matrix A, where $B = P^{-1}AP$, respectively, are

A. $\{\lambda_k \det(A) \mid 1 \leq k \leq n\}$ and $\{Pv_k \mid 1 \leq k \leq n\}$

B. $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{v_k \mid 1 \leq k \leq n\}$

C. $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{Pv_k \mid 1 \leq k \leq n\}$

D. $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{P^{-1}v_k \mid 1 \leq k \leq n\}$

Q1.23.2 MCQ 2023 2M

Let $\boldsymbol{x}$ be an $n \times 1$ real column vector with length $l = \sqrt{x^T x}$. The trace of the matrix $P = xx^T$ is

A. l^2 **B.** $\frac{l^2}{4}$ **C.** l **D.** $\frac{l^2}{2}$

Q1.22.1 MCQ 2022 1M

Consider a system of linear equations $Ax = b$, where

$$A = \begin{bmatrix} 1 & -\sqrt{2} & 3 \\ -1 & \sqrt{2} & -3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

This system of equations admits _____.

A. a unique solution for x

B. infinitely many solutions for x

C. no solutions for x

D. exactly two solutions for x

Q1.22.2 MCQ 2022 2M

Let α, β be two non-zero real numbers and v_1, v_2 be two non-zero real vectors of size 3×1. Suppose that v_1 and v_2 satisfy $v_1^T v_2 = 0$, $v_1^T v_1 = 1$, and $v_2^T v_2 = 1$.
Let A be the 3×3 matrix given by:

$$A = \alpha v_1 v_1^T + \beta v_2 v_2^T$$

The eigenvalues of A are _____.

A. $0, \alpha, \beta$

B. $0, \alpha + \beta, \alpha - \beta$

C. $0, \frac{\alpha + \beta}{2}, \sqrt{\alpha\beta}$

D. $0, 0, \sqrt{\alpha^2 + \beta^2}$

Q1.21.1 NAT 2021 1M

If the vectors $(1.0, -1.0, 2.0)$, $(7.0, 3.0, x)$ and $(2.0, 3.0, 1.0)$ in $\mathbb{R}^3$ are linearly dependent, the value of x is _____

Q1.21.2 NAT 2021 2M

A real 2×2 non-singular matrix A with repeated eigenvalue is given as

$$\mathbf{A} = \begin{bmatrix} x & -3.0 \\ 3.0 & 4.0 \end{bmatrix}$$

where x is a real positive number. The value of x (rounded off to one decimal place) is _____

Q1.20.1 MCQ 2020 1M

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_6$ are six vectors in $\mathbb{R}^4$, which one of the following statements is FALSE?

A. It is not necessary that these vectors span $\mathbb{R}^4$.

B. These vectors are not linearly independent.

C. Any four of these vectors form a basis for $\mathbb{R}^4$.

D. If $\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5, \mathbf{v}_6\}$ spans $\mathbb{R}^4$, then it forms a basis for $\mathbb{R}^4$.

Q1.20.2 MCQ 2020 2M

Consider the following system of linear equations.

$$x_1 + 2x_2 = b_1; \quad 2x_1 + 4x_2 = b_2;$$
$$3x_1 + 7x_2 = b_3; \quad 3x_1 + 9x_2 = b_4$$

Which one of the following conditions ensures that a solution exists for the above system?

A. $b_2 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

B. $b_3 = 2b_1$ and $6b_1 - 3b_3 + b_4 = 0$

C. $b_2 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

D. $b_3 = 2b_1$ and $3b_1 - 6b_3 + b_4 = 0$

Q1.19.1 NAT 2019 1M

The number of distinct eigenvalues of the matrix

$$A = \begin{bmatrix} 2 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

is equal to _____.

Q1.18.1 MCQ 2018 1M

Let M be a real 4×4 matrix. Consider the following statements:

S1: M has 4 linearly independent eigenvectors.

S2: M has 4 distinct eigenvalues.

S3: M is non-singular (invertible).

Which one among the following is TRUE?

A. S1 implies S2 **B.** S1 implies S3
C. S2 implies S1 **D.** S3 implies S2

Q1.18.2 NAT 2018 1M

Consider matrix $A = \begin{bmatrix} k & 2k \\ k^2 - k & k^2 \end{bmatrix}$ and vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. The number of distinct real values of k for which the equation $Ax = 0$ has infinitely many solutions is _____.

Q1.17.1 NAT 2017 1M

The rank of the matrix

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

is _____.

Q1.17.2 MCQ 2017 1M

Consider the 5×5 matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 1 & 2 \\ 2 & 3 & 4 & 5 & 1 \end{bmatrix}.$$

It is given that A has only one real eigenvalue. Then the real eigenvalue of A is

A. -2.5 **B.** 0 **C.** 15 **D.** 25

Q1.17.3 MCQ 2017 1M

The rank of the matrix $M = \begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.16.1 MCQ 2016 1M

Let $M^4 = I$ (where I denotes the identity matrix) and $M \neq I$, $M^2 \neq I$ and $M^3 \neq I$. Then, for any natural number k, M^{-1} equals:

A. M^{4k+1} **B.** M^{4k+2} **C.** M^{4k+3} **D.** M^{4k}

Q1.16.2 NAT 2016 1M

The value of x for which the matrix

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 9 & 7 & 13 \\ -6 & -4 & -9 + x \end{bmatrix}$$

has zero as an eigenvalue is _____.

Q1.16.3 NAT 2016 2M

The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has $\det(A) = 100$ and $\text{trace}(A) = 14$. The value of $|a - b|$ is _____.

Q1.16.4 MCQ 2016 1M

Consider a 2×2 square matrix

$$\mathbf{A} = \begin{bmatrix} \sigma & x \\ \omega & \sigma \end{bmatrix},$$

where x is unknown. If the eigenvalues of the matrix $\mathbf{A}$ are $(\sigma + j\omega)$ and $(\sigma - j\omega)$, then x is equal to

A. $+j\omega$ **B.** $-j\omega$ **C.** $+\omega$ **D.** $-\omega$

Q1.16.5 MCQ 2016 2M

If the vectors $e_1 = (1, 0, 2)$, $e_2 = (0, 1, 0)$ and $e_3 = (-2, 0, 1)$ form an orthogonal basis of the three-dimensional real space $\mathbb{R}^3$, then the vector $u = (4, 3, -3) \in \mathbb{R}^3$ can be expressed as

A. $u = -\frac{2}{5}e_1 - 3e_2 - \frac{11}{5}e_3$

B. $u = -\frac{2}{5}e_1 - 3e_2 + \frac{11}{5}e_3$

C. $u = -\frac{2}{5}e_1 + 3e_2 + \frac{11}{5}e_3$

D. $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

Q1.15.1 NAT 2015 1M

Consider a system of linear equations:

$$x - 2y + 3z = -1,$$
$$x - 3y + 4z = 1, \text{ and}$$
$$-2x + 4y - 6z = k.$$

The value of k for which the system has infinitely many solutions is _____.

Q1.15.2 NAT 2015 1M

The value of p such that the vector $\begin{bmatrix}1\\2\\3\end{bmatrix}$ is an eigenvector of the matrix $\begin{bmatrix}4 & 1 & 2\\ p & 2 & 1\\ 14 & -4 & 10\end{bmatrix}$ is _____.

Q1.15.3 MCQ 2015 1M

The value of x for which all the eigen-values of the matrix given below are real is

$$\begin{bmatrix}10 & 5+j & 4\\ x & 20 & 2\\ 4 & 2 & -10\end{bmatrix}$$

A. $5+j$ **B.** $5-j$ **C.** $1-5j$ **D.** $1+5j$

Q1.15.4 MCQ 2015 1M

For $A = \begin{bmatrix}1 & \tan x\\ -\tan x & 1\end{bmatrix}$, the determinant of $A^T A^{-1}$ is

A. $\sec^2 x$ **B.** $\cos 4x$ **C.** 1 **D.** 0

Q1.14.1 NAT 2014 1M

A real (4×4) matrix A satisfies the equation $A^2 = I$, where I is the (4×4) identity matrix. The positive eigen value of A is _____.

Q1.14.2 MCQ 2014 1M

For matrices of same dimension M, N and scalar c, which one of these properties DOES NOT ALWAYS hold?

A. $(M^T)^T = M$ **B.** $(cM)^T = c(M)^T$

C. $(M+N)^T = M^T + N^T$ **D.** $MN = NM$

Q1.14.3 NAT 2014 2M

Consider the matrix

$$J_6 = \begin{bmatrix}0&0&0&0&0&1\\0&0&0&0&1&0\\0&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&0&0&0\\1&0&0&0&0&0\end{bmatrix}$$

which is obtained by reversing the order of the columns of the identity matrix I_6. Let $P = I_6 + \alpha J_6$, where α is a non-negative real number. The value of α for which $\det(P) = 0$ is _____.

Q1.14.4 NAT 2014 1M

The determinant of matrix A is 5 and the determinant of matrix B is 40. The determinant of matrix AB is _____.

Q1.14.5 MCQ 2014 2M

The system of linear equations

$$\begin{pmatrix}2&1&3\\3&0&1\\1&2&5\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix} = \begin{pmatrix}5\\-4\\14\end{pmatrix} \text{ has}$$

A. a unique solution

B. infinitely many solutions

C. no solution

D. exactly two solutions

Q1.14.6 NAT 2014 2M

The maximum value of the determinant among all 2×2 real symmetric matrices with trace 14 is _____.

Q1.14.7 MCQ 2014 2M

Which one of the following statements is NOT true for a square matrix A?

A. If A is upper triangular, the eigenvalues of A are the diagonal elements of it

B. If A is real symmetric, the eigenvalues of A are always real and positive

C. If A is real, the eigenvalues of A and A^T are always the same

D. If all the principal minors of A are positive, all the eigenvalues of A are also positive

Q1.13.1 MCQ 2013 1M

The minimum eigenvalue of the following matrix is

$$\begin{bmatrix}3&5&2\\5&12&7\\2&7&5\end{bmatrix}$$

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.13.2 MCQ 2013 2M

Let A be an $m \times n$ matrix and B an $n \times m$ matrix. It is given that determinant $(I_m + AB)$ = determinant $(I_n + BA)$, where I_k is the $k \times k$ identity matrix. Using the above property, the determinant of the matrix given below is

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

A. 2 **B.** 5 **C.** 8 **D.** 16

Q1.12.1 MCQ 2012 2M

Given that

$$A = \begin{bmatrix} -5 & -3 \\ 2 & 0 \end{bmatrix} \text{ and } I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ the value of } A^3 \text{ is}$$

A. $15A + 12I$ **B.** $19A + 30I$

C. $17A + 15I$ **D.** $17A + 21I$

Q1.11.1 MCQ 2011 2M

The system of equations

$$x + y + z = 6$$
$$x + 4y + 6z = 20$$
$$x + 4y + \lambda z = \mu$$

has **NO** solution for values of λ and μ given by

A. $\lambda = 6, \mu = 20$ **B.** $\lambda = 6, \mu \neq 20$

C. $\lambda \neq 6, \mu = 20$ **D.** $\lambda \neq 6, \mu \neq 20$

Q1.10.1 MCQ 2010 1M

The eigenvalues of a skew-symmetric matrix are

A. always zero

B. always pure imaginary

C. either zero or pure imaginary

D. always real

Q1.09.1 MCQ 2009 2M

The eigen values of the following matrix are

$$\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

A. $3, 3 + 5j, 6 - j$ **B.** $-6 + 5j, 3 + j, 3 - j$

C. $3 + j, 3 - j, 5 + j$ **D.** $3, -1 + 3j, -1 - 3j$

Q1.08.1 MCQ 2008 1M

All the four entries of the 2×2 matrix $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$ are nonzero, and one of its eigenvalues is zero. Which of the following statements is true?

A. $p_{11}p_{22} - p_{12}p_{21} = 1$ **B.** $p_{11}p_{22} - p_{12}p_{21} = -1$

C. $p_{11}p_{22} - p_{12}p_{21} = 0$ **D.** $p_{11}p_{22} + p_{12}p_{21} = 0$

Q1.08.2 MCQ 2008 1M

The system of linear equations

$$4x + 2y = 7$$

$$2x + y = 6$$

has

A. a unique solution

B. no solution

C. an infinite number of solutions

D. exactly two distinct solutions

Q1.06.1 MCQ 2006 1M

The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is:

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.06.2 MCQ 2006 2M

The eigenvalues and the corresponding eigenvectors of a 2×2 matrix are given by

Eigenvalue	Eigenvector
$\lambda_1 = 8$	$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
$\lambda_2 = 4$	$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is:

A. $\begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$ **B.** $\begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$ **C.** $\begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$ **D.** $\begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$

Q1.06.3 MCQ 2006 2M

For the matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ the eigenvalue corresponding to the eigenvector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$ is:

A. 2 **B.** 4 **C.** 6 **D.** 8

Q1.05.1 MCQ 2005 2M

Given the matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$, $[AA^T]^{-1}$ is:

A. $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ **B.** $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}$

Q1.05.2 MCQ 2005 2M

Given the matrix $\begin{bmatrix} -4 & 2 \\ 4 & 3 \end{bmatrix}$, the eigen vector is

A. $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ **B.** $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$ **C.** $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ **D.** $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Q1.00.1 MCQ 2000 1M

The eigen values of the matrix $\begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & -1 & 4 \end{bmatrix}$ are

A. 2, -2, 1, -1 **B.** 2, 3, -2, 4
C. 2, 3, 1, 4 **D.** None of the above

Q1.98.1 MCQ 1998 1M

The eigen values of the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ are

A. 1, 1 **B.** -1, -1 **C.** $j, -j$ **D.** 1, -1

Q1.94.1 MCQ 1994 1M

The rank of $(m \times n)$ matrix $(m < n)$ cannot be more than

A. m **B.** n **C.** mn **D.** None

Q1.94.2 MCQ 1994 2M

The following system of equations

$$x_1 + x_2 + x_3 = 3$$
$$x_1 - x_3 = 0$$
$$x_1 - x_2 + x_3 = 1$$

has

A. Unique solution

B. No solution

C. Infinite number of solutions

D. Only one solution

Electrical Engineering

Q1.26.1 MSQ 2026 1M

Two $n \times n$ matrices A and B have a common eigenvalue 2, and the same corresponding nonzero eigenvector.
Which of the following options is/are correct?
(Note: I is the $n \times n$ identity matrix.)

☐ **A.** Determinant $(A - 2I) = 0$

☐ **B.** Determinant $(B - 2I) = 0$

☐ **C.** Determinant $(A + B - 2I) = 0$

☐ **D.** Determinant $(A + B - 4I) = 0$

Q1.26.2 NAT 2026 1M

A is an $m \times m$ skew-symmetric matrix with real-valued entries, and x is an m-dimensional column vector with real-valued entries such that $x^{\mathrm{T}}x = 1$. The quantity $x^{\mathrm{T}}Ax$ evaluates to _____. (Answer in integer)

Q1.26.3 MCQ 2026 2M

Which one of the following statements is ALWAYS correct about a collection of p column vectors, each having n real-valued entries?

A. If $p > n$, then the column vectors must be linearly dependent

B. If $p > n$, then the column vectors must be linearly independent

C. If $p = n$, then the column vectors must be orthogonal

D. If $p < n$, then the column vectors must be linearly independent

Q1.26.4 MSQ 2026 2M

Consider an $n \times n$ orthogonal matrix A with real entries and each column having unit Euclidean norm. Which of the following statements is/are correct?

☐ **A.** The value of the determinant of A is either $+1$ or -1

☐ **B.** The eigenvalues of A have modulus 1

☐ **C.** $\|Ax\| = \|x\|$, for all $x \in R^n$, where $\|x\|$ denotes the Euclidean norm of x, and $(Ax)^T(Ay) \neq x^T y$, for all distinct $x, y \in R^n$

☐ **D.** $\|Ax\| = \|x\|$, for all $x \in R^n$, where $\|x\|$ denotes the Euclidean norm of x, and $(Ax)^T(Ay) = x^T y$, for all $x, y \in R^n$

Q1.26.5 MSQ 2026 2M

Consider the system of linear equations: $Ax = b$, where A is an $n \times n$ matrix, and x and b are n-dimensional column vectors. Suppose this system of equations has a unique solution. Which of the following statements is/are correct?

☐ **A.** A^{-1} exists

☐ **B.** The system of equations $A^m x = b$ also has a unique solution for $m = 1, 2, 3, \ldots$

☐ **C.** $\text{rank}(A) = \text{rank}(A^m)$, for $m = 1, 2, 3, \ldots$

☐ **D.** $\text{rank}(A) < \text{rank}([A \mid b])$, where $[A \mid b]$ denotes the augmented matrix.

Q1.25.1 MCQ 2025 1M

Let v_1 and v_2 be the two eigenvectors corresponding to distinct eigenvalues of a 3×3 real symmetric matrix. Which one of the following statements is true?

A. $v_1^T v_2 \neq 0$ **B.** $v_1^T v_2 = 0$

C. $v_1 + v_2 = 0$ **D.** $v_1 - v_2 = 0$

Q1.25.2 MCQ 2025 1M

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$, and $b = \begin{bmatrix} 1/3 \\ -1/3 \\ 0 \end{bmatrix}$. Then, the system of linear equations $Ax = b$ has

A. a unique solution.

B. infinitely many solutions.

C. a finite number of solutions.

D. no solution.

Q1.25.3 MCQ 2025 1M

Let $P = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and let I be the identity matrix.

Then P^2 is equal to

A. $2P - I$ **B.** P **C.** I **D.** $P + I$

Q1.24.1 MCQ 2024 1M

Which one of the following matrices has an inverse?

A. $\begin{bmatrix} 1 & 4 & 8 \\ 0 & 4 & 2 \\ 0.5 & 2 & 4 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 2 & 9 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 4 & 8 \\ 0 & 4 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 4 & 8 \\ 0 & 4 & 2 \\ 3 & 12 & 24 \end{bmatrix}$

Q1.24.2 NAT 2024 1M

The sum of the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^2$ is _____ .

(rounded off to the nearest integer)

Q1.23.1 MCQ 2023 1M

In the figure, the vectors u and v are related as: $Au = v$ by a transformation matrix A. The correct choice of A is

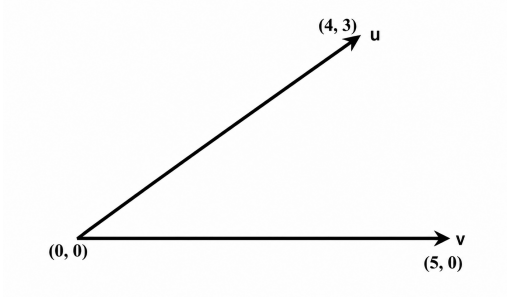


Fig. Q1.23.1

A. $\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{bmatrix}$ **B.** $\begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$

C. $\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$ **D.** $\begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & -\frac{4}{5} \end{bmatrix}$

Q1.22.1 MCQ 2022 1M

Consider a 3×3 matrix A whose (i, j)-th element, $a_{i,j} = (i - j)^3$. Then the matrix A will be

A. symmetric **B.** skew-symmetric

C. unitary **D.** null

Q1.22.2 MCQ 2022 2M

e^A denotes the exponential of a square matrix A. Suppose λ is an eigenvalue and v is the corresponding eigen-vector of matrix A. Consider the following two statements:

Statement 1: e^λ is an eigenvalue of e^A.

Statement 2: v is an eigen-vector of e^A.

Which one of the following options is correct?

A. Statement 1 is true and statement 2 is false.

B. Statement 1 is false and statement 2 is true.

C. Both the statements are correct.

D. Both the statements are false.

Q1.22.3 MCQ 2022 2M

Consider a matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -2 \\ 0 & 1 & 1 \end{bmatrix}$. The matrix A satisfies the equation $6A^{-1} = A^2 + cA + dI$, where c and d are scalars and I is the identity matrix. Then $(c + d)$ is equal to

A. 5 **B.** 17 **C.** -6 **D.** 11

Q1.21.1 MCQ 2021 1M

Let p and q be real numbers such that $p^2 + q^2 = 1$. The eigenvalues of the matrix

$$\begin{bmatrix} p & q \\ q & -p \end{bmatrix}$$

are

A. 1 and 1 **B.** 1 and -1

C. j and $-j$ **D.** pq and $-pq$

Q1.21.2 NAT 2021 2M

Let A be a 10×10 matrix such that A^5 is a null matrix, and let I be the 10×10 identity matrix. The determinant of $A + I$ is _____.

Q1.20.1 MCQ 2020 2M

The number of purely real elements in a lower triangular representation of the given 3×3 matrix, obtained through the given decomposition is _____.

$$\begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 1 \\ 3 & 1 & 7 \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{12} & a_{22} & 0 \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} & 0 & 0 \\ a_{12} & a_{22} & 0 \\ a_{13} & a_{23} & a_{33} \end{bmatrix}^T$$

A. 5 **B.** 6 **C.** 8 **D.** 9

Q1.19.1 MCQ 2019 1M

M is a 2×2 matrix with eigenvalues 4 and 9. The eigenvalues of M^2 are

A. 4 and 9 **B.** 2 and 3 **C.** -2 and -3 **D.** 16 and 81

Q1.19.2 MCQ 2019 2M

Consider a 2×2 matrix $M = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$, where, v_1 and v_2 are the column vectors. Suppose $M^{-1} = \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix}$, where u_1^T and u_2^T are the row vectors. Consider the following statements:

Statement 1: $u_1^T v_1 = 1$ and $u_2^T v_2 = 1$

Statement 2: $u_1^T v_2 = 0$ and $u_2^T v_1 = 0$

Which of the following options is correct?

A. Statement 1 is true and statement 2 is false

B. Statement 2 is true and statement 1 is false

C. Both the statements are true

D. Both the statements are false

Q1.18.1 NAT 2018 1M

Consider a non-singular 2×2 square matrix A. If $\text{trace}(A) = 4$ and $\text{trace}(A^2) = 5$, the determinant of the matrix A is _____ (up to 1 decimal place).

Q1.18.2 NAT 2018 2M

Let $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ and $B = A^3 - A^2 - 4A + 5I$, where I is the 3×3 identity matrix. The determinant of B is _____ (up to 1 decimal place).

Q1.17.1 MCQ 2017 1M

The matrix $A = \begin{bmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{bmatrix}$ has three distinct eigenvalues and one of its eigenvectors is $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Which one of the following can be another eigenvector of A?

A. $\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$ **B.** $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ **C.** $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Q1.17.2 MCQ 2017 2M

The eigenvalues of the matrix given below are

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix}$$

A. (0,-1,-3) **B.** (0, -2, -3) **C.** (0, 2, 3) **D.** (0, 1, 3)

Q1.16.1 NAT 2016 1M

Consider a 3×3 matrix with every element being equal to 1. Its only non-zero eigenvalue is _____.

Q1.16.2 MCQ 2016 2M

Let the eigenvalues of a 2×2 matrix A be $1, -2$ with eigenvectors x_1 and x_2 respectively. Then the eigenvalues and eigenvectors of the matrix $A^2 - 3A + 4I$ would, respectively, be

A. 2, 14; x_1, x_2 **B.** 2, 14; $x_1 + x_2, x_1 - x_2$

C. 2, 0; x_1, x_2 **D.** 2, 0; $x_1 + x_2, x_1 - x_2$

Q1.16.3 MCQ 2016 2M

Let A be a 4×3 real matrix with rank 2. Which one of the following statement is TRUE?

A. Rank of $A^T A$ is less than 2.

B. Rank of $A^T A$ is equal to 2.

C. Rank of $A^T A$ is greater than 2.

D. Rank of $A^T A$ can be any number between 1 and 3.

Q1.16.4 MCQ 2016 1M

A 3×3 matrix P is such that, $P^3 = P$. Then the eigenvalues of P are

A. $1, 1, -1$

B. $1, 0.5 + j0.866, 0.5 - j0.866$

C. $1, -0.5 + j0.866, -0.5 - j0.866$

D. $0, 1, -1$

Q1.16.5 MCQ 2016 2M

Let $P = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Consider the set S of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ such that $a^2 + b^2 = 1$ where $\begin{pmatrix} a \\ b \end{pmatrix} = P \begin{pmatrix} x \\ y \end{pmatrix}$. Then S is

A. a circle of radius $\sqrt{10}$

B. a circle of radius $\frac{1}{\sqrt{10}}$

C. an ellipse with major axis along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

D. an ellipse with minor axis along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q1.15.1 NAT 2015 1M

If the sum of the diagonal elements of a 2×2 matrix is -6, then the maximum possible value of determinant of the matrix is _____.

Q1.15.2 MCQ 2015 2M

The maximum value of "a" such that the matrix

$$\begin{pmatrix} -3 & 0 & -2 \\ 1 & -1 & 0 \\ 0 & a & -2 \end{pmatrix}$$

has three linearly independent real eigenvectors is

A. $\frac{2}{3\sqrt{3}}$ **B.** $\frac{1}{3\sqrt{3}}$ **C.** $\frac{1+2\sqrt{3}}{3\sqrt{3}}$ **D.** $\frac{1+\sqrt{3}}{3\sqrt{3}}$

Q1.15.3 MCQ 2015 1M

We have a set of 3 linear equations in 3 unknowns. '$X \equiv Y$' means X and Y are equivalent statements and '$X \not\equiv Y$' means X and Y are not equivalent statements.

P: There is a unique solution.

Q: The equations are linearly independent.

R: All eigenvalues of the coefficient matrix are nonzero.

S: The determinant of the coefficient matrix is nonzero.

Which one of the following is TRUE?

A. $P \equiv Q \equiv R \equiv S$ **B.** $P \equiv R \not\equiv Q \equiv S$

C. $P \equiv Q \not\equiv R \equiv S$ **D.** $P \not\equiv Q \not\equiv R \not\equiv S$

Q1.14.1 NAT 2014 2M

A system matrix is given as follows

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -6 & -11 & 6 \\ -6 & -11 & 5 \end{bmatrix}.$$

The absolute value of the ratio of the maximum eigenvalue to the minimum eigenvalue is _____

Q1.14.2 MCQ 2014 1M

Two matrices A and B are given below:

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}; \qquad B = \begin{bmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{bmatrix}$$

If the rank of matrix A is N, then the rank of matrix B is

A. $N/2$ **B.** $N - 1$ **C.** N **D.** $2N$

Q1.14.3 MCQ 2014 1M

Which one of the following statements is true for all real symmetric matrices?

A. All the eigenvalues are real.

B. All the eigenvalues are positive.

C. All the eigenvalues are distinct.

D. Sum of all the eigenvalues is zero.

Q1.13.1 MCQ 2013 2M

A matrix has eigenvalues -1 and -2. The corresponding eigenvectors are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ respectively. The matrix is

A. $\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

Q1.11.1 MCQ 2011 1M

The two vectors $[1, 1, 1]$ and $[1, a, a^2]$, where $a = \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$, are

A. Orthonormal **B.** Orthogonal

C. Parallel **D.** Collinear

Q1.11.2 MCQ 2011 2M

The matrix $[A] = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}$ is decomposed into a product of a lower triangular matrix $[L]$ and an upper triangular matrix $[U]$. The properly decomposed $[L]$ and $[U]$ matrices respectively are

A. $\begin{bmatrix} 1 & 0 \\ 4 & -1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$ **B.** $\begin{bmatrix} 2 & 0 \\ 4 & -1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$

Q1.10.1 MCQ 2010 2M

An eigenvector of $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ is

A. $[-1 \;\; 1 \;\; 1]^\top$ **B.** $[1 \;\; 2 \;\; 1]^\top$

C. $[1 \;\; -1 \;\; 2]^\top$ **D.** $[2 \;\; 1 \;\; -1]^\top$

Q1.09.1 MCQ 2009 1M

The trace and determinant of a 2×2 matrix are known to be -2 and -35 respectively. Its eigenvalues are

A. -30 and -5 **B.** -37 and -1

C. -7 and 5 **D.** 17.5 and -2

Q1.08.1 MCQ 2008 1M

The characteristic equation of a (3×3) matrix P is defined as

$$\alpha(\lambda) = |\lambda I - P| = \lambda^3 + \lambda^2 + 2\lambda + 1 = 0.$$

If I denotes identity matrix, then the inverse of matrix P will be

A. $(P^2 + P + 2I)$ **B.** $(P^2 + P + I)$

C. $-(P^2 + P + I)$ **D.** $-(P^2 + P + 2I)$

Q1.08.2 MCQ 2008 1M

If the rank of a (5×6) matrix Q is 4, then which one of the following statements is correct?

A. Q will have four linearly independent rows and four linearly independent columns

B. Q will have four linearly independent rows and five linearly independent columns

C. $QQ^\top$ will be invertible

D. $Q^\top Q$ will be invertible

Q1.08.3 MCQ 2008 2M

Let P be a 2×2 real orthogonal matrix and $\vec{x}$ is a real vector $[x_1, x_2]^\top$ with length $\|\vec{x}\| = \left(x_1^2 + x_2^2\right)^{\frac{1}{2}}$. Then, which one of the following statements is correct?

A. $\|P\vec{x}\| \leq \|\vec{x}\|$ where at least one vector satisfies $\|P\vec{x}\| < \|\vec{x}\|$

B. $\|P\vec{x}\| = \|\vec{x}\|$ for all vectors $\vec{x}$

C. $\|P\vec{x}\| \geq \|\vec{x}\|$ where at least one vector satisfies $\|P\vec{x}\| > \|\vec{x}\|$

D. No relationship can be established between $\|\vec{x}\|$ and $\|P\vec{x}\|$

Q1.08.4 MCQ 2008 2M

A is a $m \times n$ full rank matrix with $m > n$ and I is an identity matrix. Let matrix $A^+ = (A^\top A)^{-1} A^\top$. Then, which one of the following statements is FALSE?

A. $AA^+A = A$ **B.** $(AA^+)^2 = AA^+$

C. $A^+A = I$ **D.** $AA^+A = A^+$

Q1.07.1 MCQ 2007 1M

$\mathbf{x} = [x_1 \;\; x_2 \;\; \ldots \;\; x_n]^\top$ is an n-tuple nonzero vector. The $n \times n$ matrix $\mathbf{V} = \mathbf{x}\mathbf{x}^\top$

A. has rank zero **B.** has rank 1

C. is orthogonal **D.** has rank n

Q1.07.2 MCQ 2007 2M

Let x and y be two vectors in a 3 dimensional space and $\langle x, y \rangle$ denote their dot product. Then the determinant

$$\det \begin{bmatrix} \langle x, x \rangle & \langle x, y \rangle \\ \langle y, x \rangle & \langle y, y \rangle \end{bmatrix}$$

A. is zero when x and y are linearly independent

B. is positive when x and y are linearly independent

C. is non-zero for all non-zero x and y

D. is zero only when either x or y is zero

Q1.07.3 MCQ 2007 2M

The linear operation $L(\mathbf{x})$ is defined by the cross product $L(\mathbf{x}) = \mathbf{b} \times \mathbf{x}$, where $\mathbf{b} = [0 \;\; 1 \;\; 0]^\top$ and $\mathbf{x} = [x_1 \;\; x_2 \;\; x_3]^\top$

are three dimensional vectors. The 3×3 matrix M of this operation satisfies

$$L(\mathbf{x}) = M \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Then the eigenvalues of M are

A. $0, +1, -1$ **B.** $1, -1, 1$ **C.** $i, -i, 1$ **D.** $i, -i, 0$

Common Data Questions

Answer Questions Q1.07.4 and Q1.07.5 using the following data. Cayley-Hamilton Theorem states that a square matrix satisfies its own characteristic equation.

Consider a matrix: $A = \begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}$

Q1.07.4 MCQ 2007 1M

A satisfies the relation

A. $A + 3I + 2A^{-1} = 0$ **B.** $A^2 + 2A + 2I = 0$

C. $(A + I)(A + 2I) = 0$ **D.** $\exp(A) = 0$

Q1.07.5 MCQ 2007 1M

For the shared matrix $A = \begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}$, A^9 equals

A. $511\,A + 510\,I$ **B.** $309\,A + 104\,I$

C. $154\,A + 155\,I$ **D.** $\exp(9\,A)$

Q1.05.1 MCQ 2005 1M

In the matrix equation $Px = q$, which of the following is a necessary condition for the existence of at least one solution for the unknown vector x:

A. Augmented matrix $[P \quad q]$ must have the same rank as matrix P

B. Vector q must have only non-zero elements

C. Matrix P must be singular

D. Matrix P must be square

Q1.05.2 MCQ 2005 2M

If $R = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{bmatrix}$, the top row of R^{-1} is:

A. $\begin{bmatrix} 5 & 6 & 4 \end{bmatrix}$ **B.** $\begin{bmatrix} 5 & -3 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 2 & 0 & -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 & -1 & \frac{1}{2} \end{bmatrix}$

Q1.02.1 MCQ 2002 1M

The determinant of the matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 100 & 1 & 0 & 0 \\ 100 & 200 & 1 & 0 \\ 100 & 200 & 300 & 1 \end{bmatrix}$ is:

A. 100 **B.** 200 **C.** 1 **D.** 300

Q1.98.1 MCQ 1998 1M

The vector $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ is an eigen vector of $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.

One of the given values of A is:

A. 1 **B.** 2 **C.** 5 **D.** -1

Q1.97.1 NAT 1997 5M

Express the given matrix A as a product of two triangular matrices, L and U, where the diagonal elements of the lower triangular matrix L are unity and U is an upper triangular matrix.

$$a = \begin{bmatrix} 2 & 1 & 5 \\ 4 & 8 & 13 \\ 6 & 27 & 31 \end{bmatrix}$$

Q1.95.1 NAT 1995 2M

A set three linear equations with unknowns X, Y and Z is shown below.

$$\begin{bmatrix} 0 & 21.432 & 34.432 \\ 17.587 & -1.348 & -1.865 \\ 0 & -2.663 & -1.803 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \begin{bmatrix} 56.042 \\ 02.631 \\ 05.952 \end{bmatrix}$$

Decompose the coefficient matrix into an upper triangular matrix and then solve for the unknowns, X, Y and Z.

Q1.94.1 MCQ 1994 1M

A 5×7 matrix has all its entries equal to -1. The rank of the matrix is

A. 7 **B.** 5 **C.** 1 **D.** Zero

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Q1.26.1 MCQ 2026 1M

A matrix $P \in \mathbb{R}^{3\times3}$ satisfies $P^3 - 4P^2 + 5P - 2I = 0$ where I is a 3×3 identity matrix. The set of all possible eigenvalues of matrix P is _____.

A. $\{1, 2, 3\}$ **B.** $\{1\}$ **C.** $\{1, 2\}$ **D.** $\{2\}$

Q1.26.2 MCQ 2026 1M

Consider a matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Let $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. The number of solutions to the linear system of equations $A\mathbf{x} = \mathbf{b}$ is _____.

A. 0 **B.** 1 **C.** 6 **D.** infinitely many

Q1.25.1 MCQ 2025 1M

A $2n \times 2n$ matrix $A = [a_{ij}]$ has its elements as

$$a_{ij} = \begin{cases} \beta & \text{if } (i+j) \text{ is odd,} \\ -\beta & \text{if } (i+j) \text{ is even,} \end{cases}$$

where n is any integer greater than 2 and β is any non-zero real number. The rank of A is

A. 1 **B.** 2 **C.** n **D.** 2n

Q1.25.2 NAT 2025 1M

If one of the eigenvectors of the matrix $A = \begin{bmatrix} -1 & -1 \\ x & -4 \end{bmatrix}$ is along the direction of $\begin{bmatrix} \alpha \\ 2\alpha \end{bmatrix}$, where α is any non-zero real number, then the value of x is _____ (in integer).

Q1.24.1 MCQ 2024 1M

A matrix M is constructed by stacking three column vectors v_1, v_2, v_3 as

$$M = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}.$$

Choose the set of vectors from the following options such that $\text{rank}(M) = 3$.

A. $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

B. $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

C. $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

D. $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$

Q1.24.2 NAT 2024 1M

A 3×3 matrix P with all real elements has eigenvalues $\frac{1}{4}$, 1, and -2. The value of $|P^{-1}|$ is _____ (rounded off to nearest integer).

Q1.23.1 MCQ 2023 1M

Choose solution set S corresponding to the systems of two equations

$$x - 2y + z = 0$$
$$x - z = 0$$

Note: $\mathscr{R}$ denotes the set of real numbers

A. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \middle| \alpha \in \mathscr{R} \right\}$

B. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha, \beta \in \mathscr{R} \right\}$

C. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \middle| \alpha, \beta \in \mathscr{R} \right\}$

D. $S = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha \in \mathcal{R} \right\}$

Q1.23.2 NAT 2023 2M

The rank of the matrix A given below is one. The ratio $\frac{\alpha}{\beta}$ is _____ (rounded off to the nearest integer).

$$A = \begin{bmatrix} 1 & 4 \\ -3 & \alpha \\ \beta & 6 \end{bmatrix}$$

Q1.22.1 MSQ 2022 1M

Given $M = \begin{bmatrix} 2 & 3 & 7 \\ 6 & 4 & 7 \\ 4 & 6 & 14 \end{bmatrix}$, which of the following statement(s) is/are correct?

- ☐ **A.** The rank of M is 2
- ☐ **B.** The rank of M is 3
- ☐ **C.** The rows of M are linearly independent
- ☐ **D.** The determinant of M is 0

Q1.22.2 MSQ 2022 2M

The matrix $A = \begin{bmatrix} 4 & 3 \\ 9 & -2 \end{bmatrix}$ has eigenvalues -5 and 7. The eigenvector(s) is/are _____

A. $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ **C.** $\begin{bmatrix} 2 \\ -6 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 \\ 8 \end{bmatrix}$

Q1.21.1 MCQ 2021 1M

Consider the row vectors $v = (1, 0)$ and $w = (2, 0)$. The rank of the matrix $M = 2v^T v + 3w^T w$, where the superscript T denotes the transpose, is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q1.21.2 NAT 2021 2M

Given $A = \begin{pmatrix} 2 & 5 \\ 0 & 3 \end{pmatrix}$. The value of the determinant $|A^4 - 5A^3 + 6A^2 + 2I| =$ _____.

Q1.20.1 MCQ 2020 1M

A set of linear equations is given in the form $Ax = b$, where A is a 2×4 matrix with real number entries and $b \neq 0$. Will it be possible to solve for x and obtain a unique solution by multiplying both left and right sides of the equation by A^T (the super script T denotes the transpose) and inverting the matrix $A^T A$? Answer is _____

A. Yes, it is always possible to get a unique solution for any 2×4 matrix A.

B. No, it is not possible to get a unique solution for any 2×4 matrix A.

C. Yes, can obtain a unique solution provided the matrix $A^T A$ is well conditioned

D. Yes, can obtain a unique solution provided the matrix A is well conditioned

Q1.20.2 MCQ 2020 2M

Consider the matrix $M = \begin{bmatrix} 1 & -1 & 0 \\ 1 & -2 & 1 \\ 0 & -1 & 1 \end{bmatrix}$. One of the eigenvectors of M is

A. $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ **C.** $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Q1.19.1 NAT 2019 1M

A 3×3 matrix has eigenvalues 1, 2 and 5. The determinant of the matrix is _____.

Q1.18.1 MCQ 2018 1M

Let N be a 3 by 3 matrix with real number entries. The matrix N is such that $N^2 = 0$. The eigen values of N are

A. 0, 0, 0 **B.** 0,0,1 **C.** 0,1,1 **D.** 1,1,1

Q1.18.2 MCQ 2018 2M

Consider the following system of linear equations:

$$3x + 2ky = -2$$
$$kx + 6y = 2$$

Here x and y are the unknowns and k is a real constant. The value of k for which there are infinite number of solutions is

A. 3 **B.** 1 **C.** -3 **D.** -6

Q1.17.1 NAT 2017 1M

If $\mathbf{v}$ is a non-zero vector of dimension 3×1, then the matrix $A = \mathbf{v}\mathbf{v}^{\mathrm{T}}$ has a rank = _____

Q1.17.2 MCQ 2017 1M

The figure shows a shape ABC and its mirror image $A_1B_1C_1$ across the horizontal axis (X-axis). The coordinate transformation matrix that maps ABC to $A_1B_1C_1$ is

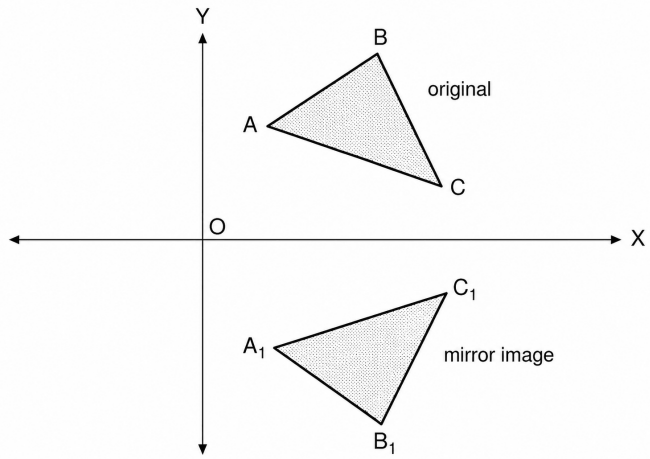


Fig. Q1.17.2

A. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **B.** $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Q1.17.3 MCQ 2017 1M

The eigenvalues of the matrix $\mathbf{A} = \begin{bmatrix} 1 & -1 & 5 \\ 0 & 5 & 6 \\ 0 & -6 & 5 \end{bmatrix}$ are

A. $-1,\ 5,\ 6$ **B.** $1,\ -5 \pm j6$

C. $1,\ 5 \pm j6$ **D.** $1,\ 5,\ 5$

Q1.16.1 NAT 2016 2M

Consider the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{pmatrix}$ whose eigenvalues are $1, -1$ and 3. Then Trace of $(\mathbf{A}^3 - 3\mathbf{A}^2)$ is _____.

Q1.15.1 MCQ 2015 1M

Let $\mathbf{A}$ be an $n \times n$ matrix with rank r $(0 < r < n)$. Then $\mathbf{Ax} = 0$ has p independent solutions, where p is

A. r **B.** n **C.** $n - r$ **D.** $n + r$

Q1.14.1 MCQ 2014 2M

A scalar valued function is defined as $f(x) = x^T Ax + b^T x + c$, where A is a symmetric positive definite matrix with dimension $n \times n$; b and x are vectors of dimension $n \times 1$. The minimum value of $f(x)$ will occur when x equals

A. $(A^T A)^{-1} b$ **B.** $-(A^T A)^{-1} b$

C. $-\left(\frac{A^{-1}b}{2}\right)$ **D.** $\frac{A^{-1}b}{2}$

Q1.13.1 MCQ 2013 1M

The dimension of the null space of the matrix $\begin{bmatrix} 0 & 1 & 1 \\ 1 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.13.2 MCQ 2013 2M

One pair of eigenvectors corresponding to the two eigenvalues of the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is

A. $\begin{bmatrix} 1 \\ -j \end{bmatrix}, \begin{bmatrix} j \\ -1 \end{bmatrix}$ **B.** $\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 \\ j \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 \\ j \end{bmatrix}, \begin{bmatrix} j \\ 1 \end{bmatrix}$

Q1.11.1 MCQ 2011 1M

The matrix $M = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ has eigenvalues $-3, -3, 5$. An eigenvector corresponding to the eigenvalue 5 is $\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}^T$. One of the eigenvectors of the matrix M^3 is

A. $\begin{bmatrix} 1 & 8 & -1 \end{bmatrix}^T$ **B.** $\begin{bmatrix} 1 & 2 & -1 \end{bmatrix}^T$

C. $\begin{bmatrix} 1 & \sqrt[3]{2} & -1 \end{bmatrix}^T$ **D.** $\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}^T$

Q1.10.1 MCQ 2010 1M

A real $n \times n$ matrix $A = [a_{ij}]$ is defined as follows:

$$a_{ij} = i, \quad \text{if } i = j$$
$$= 0, \quad \text{otherwise}$$

The summation of all n eigenvalues of A is

A. $n(n+1)/2$ **B.** $n(n-1)/2$

C. $\dfrac{n(n+1)(2n+1)}{6}$ **D.** n^2

Q1.10.2 MCQ 2010 2M

X and Y are non-zero square matrices of size $n \times n$. If $XY = 0_{n \times n}$, then

A. $|X| = 0$ and $|Y| \neq 0$ **B.** $|X| \neq 0$ and $|Y| = 0$

C. $|X| = 0$ and $|Y| = 0$ **D.** $|X| \neq 0$ and $|Y| \neq 0$

Q1.09.1 MCQ 2009 1M

Let $P \neq 0$ be a 3×3 real matrix. There exist linearly independent vectors x and y such that $Px = 0$ and $Py = 0$. The dimension of the range space of P is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.09.2 MCQ 2009 2M

The eigenvalues of a (2×2) matrix X are -2 and -3. The eigenvalues of the matrix $(X + I)^{-1}(X + 5I)$ are

A. $-3, -4$ **B.** $-1, -2$ **C.** $-1, -3$ **D.** $-2, -4$

Q1.09.3 MCQ 2009 2M

The matrix $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ rotates a vector about the axis $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ by an angle of

A. 30° **B.** 60° **C.** 90° **D.** 120°

Q1.07.1 MCQ 2007 1M

Let A be an $n \times n$ real matrix such that $A^2 = I$ and y be an n-dimensional vector. Then the linear system of equations $Ax = y$ has

A. no solution

B. a unique solution

C. more than one but finitely many independent solutions

D. infinitely many independent solutions

Q1.07.2 MCQ 2007 2M

Let $A = [a_{ij}], 1 \leq i, j \leq n$, with $n \geq 3$ and $a_{ij} = i.j$. Then the rank of A is

A. 0 **B.** 1 **C.** $n - 1$ **D.** n

Q1.06.1 MCQ 2006 1M

For a given 2×2 matrix A, it is observed that $A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $A \begin{bmatrix} 1 \\ -2 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ then the matrix A is

A. $A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$

B. $A = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

C. $A = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$

D. $A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}$

Q1.05.1 MCQ 2005 1M

Let A be a 3×3 matrix with rank 2. Then $AX = 0$ has

A. only the trivial solution $X = 0$

B. one independent solution

C. two independent solutions

D. three independent solutions

Q1.01.1 MCQ 2001 1M

The necessary condition to diagonalize a matrix is that

A. all its eigen values should be distinct

B. its eigen vectors should be independent

C. its eigen values should be real

D. the matrix is non-singular

Computer Science & IT

Q1.26.1 MCQ 2026 1M

Consider 4×4 matrices with their elements from $\{0, 1\}$. The number of such matrices with even number of 1s in every row and every column is

A. 512 **B.** 1025 **C.** 1023 **D.** 255

Q1.26.2 MCQ 2026 1M

For $n > 1$, the maximum multiplicity of any eigenvalue of an $n \times n$ matrix with elements from $\mathbb{R}$ is

A. n **B.** n - 1 **C.** 1 **D.** n + 1

Q1.26.3 MSQ 2026 1M

Let $n > 1$. Consider an $n \times n$ matrix M with its elements from $\mathbb{R}$. Let the vector $(0, 1, 0, 0, \ldots, 0) \in \mathbb{R}^n$ be in the null space of M. Which of the following options is/are always correct?

- ☐ **A.** Determinant of M is 1
- ☐ **B.** Determinant of M is 0
- ☐ **C.** Rank of M is 1
- ☐ **D.** There are at least two non-zero vectors in the null space of M

Q1.26.4 NAT 2026 1M

Consider the system of linear equations given below.

$$ax + y = b$$

$$16x + ay = 24$$

Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is _____. (*answer in integer*)

Q1.26.5 NAT 2026 2M

The determinant of a 4×4 matrix A is 3. The value of the determinant of $2A$ is _____. (*answer in integer*)

Q1.25.1 MSQ 2025 1M

Consider the given system of linear equations for variables x and y, where k is a real-valued constant. Which of the following option(s) is/are CORRECT?

$$x + ky = 1$$

$$kx + y = -1$$

A. There is exactly one value of k for which the above system of equations has no solution.

B. There exist an infinite number of values of k for which the system of equations has no solution.

C. There exists exactly one value of k for which the system of equations has exactly one solution.

D. There exists exactly one value of k for which the system of equations has an infinite number of solutions.

Q1.25.2 MCQ 2025 2M

Let A be a 2×2 matrix as given.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13} ?

A. $1, -1$ **B.** $2\sqrt{2}, -2\sqrt{2}$
C. $4\sqrt{2}, -4\sqrt{2}$ **D.** $64\sqrt{2}, -64\sqrt{2}$

Q1.25.3 MCQ 2025 1M

If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?

A. $\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$ **B.** $\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$

C. $\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$ **D.** $\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$

Q1.25.4 MCQ 2025 1M

Let L, M, and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.25.5 MSQ 2025 2M

Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3\times3}$ and $Q \in \mathbb{R}^{3\times1}$. Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Which of the following statement(s) is/are TRUE?

- **A.** The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
- **B.** If P is invertible, then both L and U are invertible.
- **C.** If P is singular, then at least one of the diagonal elements of U is zero.
- **D.** If P is symmetric, then both L and U are symmetric.

Q1.24.1 MCQ 2024 1M

The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

A. -1 **B.** 0 **C.** 1 **D.** 2

Q1.24.2 MSQ 2024 2M

Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

- **A.** There exist at least $m - n$ linearly independent solutions to this system
- **B.** There exist $m-n$ linearly independent vectors such that every solution is a linear combination of these vectors
- **C.** There exists a non-zero solution in which at least $m - n$ variables are 0
- **D.** There exists a solution in which at least n variables are non-zero

Q1.24.3 MSQ 2024 2M

Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- **A.** The determinant of B is the negative of the determinant of A
- **B.** If A is invertible, then B is also invertible
- **C.** If A is symmetric, then B is also symmetric
- **D.** If the trace of A is zero, then the trace of B is also zero

Q1.23.1 MCQ 2023 1M

Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$.

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?

- **A.** $\det(A) = \det(B)$
- **B.** $\det(B) = -\det(A)$
- **C.** $\det(A) = 0$
- **D.** $\det(AB) = \det(A) + \det(B)$

Q1.22.1 MCQ 2022 2M

Consider solving the following system of simultaneous equations using LU decomposition.

$$x_1 + x_2 - 2x_3 = 4$$

$$x_1 + 3x_2 - x_3 = 7$$

$$2x_1 + x_2 - 5x_3 = 7$$

where L and U are denoted as

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, \quad U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32}, U_{33}, and x_1 ?

- **A.** $L_{32} = 2, U_{33} = -\frac{1}{2}, x_1 = -1$
- **B.** $L_{32} = 2, U_{33} = 2, x_1 = -1$
- **C.** $L_{32} = -\frac{1}{2}, U_{33} = 2, x_1 = 0$
- **D.** $L_{32} = -\frac{1}{2}, U_{33} = -\frac{1}{2}, x_1 = 0$

Q1.22.2 MSQ 2022 2M

Which of the following is/are the eigenvector(s) for the matrix given below?

$$\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$$

A. $\begin{pmatrix}-1\\1\\0\\1\end{pmatrix}$ **B.** $\begin{pmatrix}1\\0\\-1\\0\end{pmatrix}$ **C.** $\begin{pmatrix}-1\\0\\2\\2\end{pmatrix}$ **D.** $\begin{pmatrix}0\\1\\-3\\0\end{pmatrix}$

Q1.21.1 NAT 2021 2M

Consider the following matrix.

$$\begin{pmatrix}0&1&1&1\\1&0&1&1\\1&1&0&1\\1&1&1&0\end{pmatrix}$$

The largest eigenvalue of the above matrix is _____.

Q1.21.2 NAT 2021 1M

Suppose that P is a 4×5 matrix such that every solution of the equation $P\mathbf{x} = \mathbf{0}$ is a scalar multiple of $\begin{bmatrix}2 & 5 & 4 & 3 & 1\end{bmatrix}^T$. The rank of P is _____.

Q1.20.1 MCQ 2020 1M

Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M, respectively. Consider the following statements.

I. $\text{rank}(AB) = \text{rank}(A)\text{rank}(B)$

II. $\det(AB) = \det(A)\det(B)$

III. $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

IV. $\det(A + B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

A. I and II only **B.** I and IV only

C. II and III only **D.** III and IV only

Q1.19.1 NAT 2019 2M

Consider the following matrix:

$$R = \begin{bmatrix}1&2&4&8\\1&3&9&27\\1&4&16&64\\1&5&25&125\end{bmatrix}$$

The absolute value of the product of Eigen values of R is _____.

Q1.18.1 NAT 2018 1M

Consider a matrix $A = uv^T$ where $u = \begin{pmatrix}1\\2\end{pmatrix}, v = \begin{pmatrix}1\\1\end{pmatrix}$. Note that v^T denotes the transpose of v. The largest eigenvalue of A is _____.

Q1.18.2 MCQ 2018 2M

Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix}1\\4\end{bmatrix}$. Consider the following statements:

(I) P does not have an inverse

(II) P has a repeated eigenvalue

(III) P cannot be diagonalized

Which one of the following options is correct?

A. Only I and III are necessarily true

B. Only II is necessarily true

C. Only I and II are necessarily true

D. Only II and III are necessarily true

Q1.17.1 MCQ 2017 1M

Let $c_1 \ldots c_n$ be scalars, not all zero, such that $\sum_{i=1}^{n} c_i a_i = 0$ where a_i are column vectors in R^n. Consider the set of linear equations $Ax = b$ where $A = a_1 \ldots a_n$ and $b = \sum_{i=1}^{n} a_i$. The set of equations has

A. a unique solution at $x = J_n$ where J_n denotes a n-dimensional vector of all 1

B. no solution

C. infinitely many solutions

D. finitely many solutions

Q1.17.2 MCQ 2017 2M

Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}^2 = 50$. Consider the following statements:

(I) One eigen value must be in $[-5, 5]$

(II) The eigen value with the largest magnitude must be strictly greater than 5.

Which of the above statements about eigen values of A is/are necessarily CORRECT?

A. Both (I) and (II) **B.** (I) only

C. (II) only **D.** Neither (I) nor (II)

Q1.17.3 NAT 2017 1M

Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices. Then the rank of $P + Q$ is _____.

Q1.17.4 NAT 2017 2M

If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____.

Q1.16.1 NAT 2016 1M

Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____.

Q1.16.2 MCQ 2016 1M

Consider the systems, each consisting of m linear equations in n variables.

I. If $m < n$, then all such systems have a solution

II. If $m > n$, then none of these systems has a solution

III. If $m = n$, then there exists a system which has a solution

Which one of the following is CORRECT?

A. I, II and III are true **B.** Only II and III are true

C. Only III is true **D.** None of them is true

Q1.16.3 NAT 2016 1M

Suppose that the eigenvalues of matrix A are $1, 2, 4$. The determinant of $(A^{-1})^T$ is _____.

Q1.16.4 NAT 2016 2M

Let A_1, A_2, A_3, and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5, respectively. The minimum number of scalar multiplications required to find the product $A_1A_2A_3A_4$ using the basic matrix multiplication method is _____.

Q1.15.1 MCQ 2015 2M

Consider the following 2×2 matrix A where two elements are unknown and are marked by a and b. The eigenvalues of this matrix are -1 and 7. What are the values of a and b?

$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}$$

A. $a = 6, b = 4$ **B.** $a = 4, b = 6$

C. $a = 3, b = 5$ **D.** $a = 5, b = 3$

Q1.15.2 MCQ 2015 1M

In the given matrix $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, one of the eigenvalues is 1. The eigenvectors corresponding to the eigenvalue 1 are

A. $\{\alpha(4, 2, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

B. $\{\alpha(-4, 2, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

C. $\{\alpha(\sqrt{2}, 0, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

D. $\{\alpha(-\sqrt{2}, 0, 1) \mid \alpha \neq 0, \alpha \in \mathbb{R}\}$

Q1.15.3 NAT 2015 1M

The larger of the two eigenvalues of the matrix $\begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ is _____.

Q1.15.4 NAT 2015 2M

Perform the following operations on the matrix

$$\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$$

(i)Add the third row to the second row.
(ii)Subtract the third column from the first column.
The determinant of the resultant matrix is _____.

Q1.15.5 MCQ 2015 2M

If the following system has non-trivial solution,

$$px + qy + rz = 0$$

$$qx + ry + pz = 0$$

$$rx + py + qz = 0,$$

then which one of the following options is TRUE?

A. $p - q + r = 0$ or $p = q = -r$

B. $p + q - r = 0$ or $p = -q = r$

C. $p + q + r = 0$ or $p = q = r$

D. $p - q + r = 0$ or $p = -q = -r$

Q1.14.1 NAT 2014 1M

The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a 4x4 symmetric positive definite matrix is _____.

Q1.14.2 NAT 2014 2M

The product of the non-zero eigenvalues of the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

is _____.

Q1.14.3 NAT 2014 1M

If V_1 and V_2 are 4-dimensional subspaces of a 6-dimensional vector space V, then the smallest possible dimension of $V_1 \cap V_2$ is _____.

Q1.14.4 MCQ 2014 1M

Which one of the following statements is TRUE about every $n \times n$ matrix with only real eigenvalues?

A. If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.

B. If the trace of the matrix is positive, all its eigenvalues are positive.

C. If the determinant of the matrix is positive, all its eigenvalues are positive.

D. If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

Q1.13.1 MCQ 2013 1M

Which one of the following does NOT equal $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$?

A. $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$

B. $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

C. $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

D. $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

Q1.12.1 MCQ 2012 1M

Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

A. 1024 and -1024 **B.** $1024\sqrt{2}$ and $-1024\sqrt{2}$

C. $4\sqrt{2}$ and $-4\sqrt{2}$ **D.** $512\sqrt{2}$ and $-512\sqrt{2}$

Q1.11.1 MCQ 2011 2M

Four matrices M_1, M_2, M_3 and M_4 of dimensions $p \times q$, $q \times r$, $r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of scalar multiplications is $pqr + rst + prt$. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is $pqr + prs + pst$.

If $p = 10$, $q = 100$, $r = 20$, $s = 5$, and $t = 80$, then the minimum number of scalar multiplications needed is

A. 248000 **B.** 44000 **C.** 19000 **D.** 25000

Q1.10.1 MCQ 2010 2M

Consider the following matrix

$$A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8, then

A. $x = 4, y = 10$ **B.** $x = 5, y = 8$

C. $x = -3, y = 9$ **D.** $x = -4, y = 10$

Q1.08.1 MCQ 2008 1M

The following system of equations

$$x_1 + x_2 + 2x_3 = 1$$

$$x_1 + 2x_2 + 3x_3 = 2$$

$$x_1 + 4x_2 + \alpha x_3 = 4$$

has a unique solution. The only possible value(s) for α is/are

A. 0 **B.** either 0 or 1

C. one of 0, 1 or -1 **D.** any real number

Q1.08.2 MCQ 2008 2M

How many of the following matrices have an eigenvalue 1?

$$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}1 & -1\\1 & 1\end{bmatrix} \text{ and } \begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix}$$

A. one **B.** two **C.** three **D.** four

Q1.07.1 MCQ 2007 2M

Let A be a 4×4 matrix with eigenvalues -5, -2, 1, 4.

Which of the following is an eigenvalue of $\begin{bmatrix}A & I\\I & A\end{bmatrix}$, where I is the 4×4 identity matrix?

A. -5 **B.** -7 **C.** 2 **D.** 1

Q1.04.1 MCQ 2004 1M

The number of different $n \times n$ symmetric matrices with each elements being either 0 or 1 is

A. 2^n **B.** 2^{n^2}

C. $2^{\frac{n^2+n}{2}}$ **D.** $2^{\frac{n^2-n}{2}}$

Q1.04.2 MCQ 2004 1M

Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. $ABCD = I$ then $B^{-1} =$

A. $D^{-1}C^{-1}A^{-1}$ **B.** CDA

C. ABC **D.** does not exist

Q1.01.1 MCQ 2001 1M

Consider the following statements:

S1: The sum of two singular $n \times n$ matrices may be non-singular

S2: The sum of two $n \times n$ non-singular matrices may be singular.

Which of the following statements is correct?

A. S1 and S2 are both true **B.** S1 is true, S2 is false

C. S1 is false, S2 is true **D.** S1 and S2 are both false

Q1.00.1 MCQ 2000 1M

An $n \times n$ array v is defined as follows:

$$v[i, j] = i - j \text{ for all } i, j, 1 \le i \le n, 1 \le j \le n$$

The sum of the elements of the array v is

A. 0 **B.** n -1 **C.** $n^2 - 3n + 2$ **D.** $n^2\frac{(n+1)}{2}$

Q1.98.1 MCQ 1998 2M

Consider the following determinant $\Delta = \begin{vmatrix}1 & a & bc\\1 & b & ca\\1 & C & ab\end{vmatrix}$. Which of the following is a factor of Δ?

A. a+b **B.** a-b **C.** a+b+c **D.** abc

Q1.97.1 MCQ 1997 1M

Let $A = (a_{ij})$ be an n-rowed square matrix and I_{12} be the matrix obtained by interchanging the first and second rows of the n-rowed Identify matrix. Then AI_{12} is such that its first

A. row is the same as its second row

B. row is the same as the second row of A

C. column is the same as the second column of A

D. row is all zero

Q1.96.1 NAT 1996 5M

Let $A = \begin{bmatrix}a_{11} & a_{12}\\a_{21} & a_{22}\end{bmatrix}$ and $B = \begin{bmatrix}b_{11} & b_{12}\\b_{21} & b_{22}\end{bmatrix}$ be two matrices such that $AB = I$. Let $C = A\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}$ and $CD = I$. Express the elements of D in terms of the elements of B.

Q1.96.2 MCQ 1996 2M

The matrices $\begin{bmatrix}\cos\theta & -\sin\theta\\\sin\theta & \cos\theta\end{bmatrix}$ and $\begin{bmatrix}a & 0\\0 & b\end{bmatrix}$ commute under multiplication

A. if $a = b$ or $\theta = n\pi$, n is an integer

B. always

C. never

D. if $a\cos\theta \neq b\sin\theta$

Q1.96.3 MCQ 1996 1M

Let $Ax = b$ be a system of linear equations where A is an $m \times n$ matrix and b is a $m \times 1$ column vector and X is a $n \times 1$ column vector of unknowns. Which of the following is false?

A. The system has a solution if and only if, both A and the augmented matrix $[A \quad b]$ have the same rank.

B. If $m < n$ and b is the zero vector, then the system has infinitely many solutions.

C. If $m = n$ and b is non-zero vector, then the system has a unique solution.

D. The system will have only a trivial solution when $m = n$, b is the zero vector and rank$(A) = n$.

Q1.95.1 MCQ 1995 1M

The rank of the following $(n + 1) \times (n + 1)$ matrix, where a is a real number is

$$\begin{bmatrix} 1 & a & a^2 & \dots & a^n \\ 1 & a & a^2 & \dots & a^n \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a & a^2 & \dots & a^n \end{bmatrix}$$

A. 1 **B.** 2 **C.** n **D.** Depends on the value of a

Q1.94.1 MCQ 1994 1M

Let A and B be real symmetric matrices of size $n \times n$. Then which one of the following is true?

A. $AA' = I$ **B.** $A = A^{-1}$

C. $AB = BA$ **D.** $(AB)' = BA$

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Q1.25.1 MCQ 2025 1M

Let A and B be real symmetric matrices of same size. Which one of the following options is correct?

A. $A^{\mathrm{T}} = A^{-1}$ **B.** $AB = BA$

C. $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$ **D.** $A = A^{-1}$

Q1.24.1 MCQ 2024 1M

Consider the system of linear equations

$$x + 2y + z = 5$$

$$2x + ay + 4z = 12$$

$$2x + 4y + 6z = b$$

The values of a and b such that there exists a non-trivial null space and the system admits infinite solutions are

A. a = 8, b = 14 **B.** a = 4, b = 12

C. a = 8, b = 12 **D.** a = 4, b = 14

Q1.24.2 MCQ 2024 2M

The matrix $\begin{bmatrix} 1 & a \\ 8 & 3 \end{bmatrix}$ (where $a > 0$) has a negative eigenvalue if a is greater than

A. $\frac{3}{8}$ **B.** $\frac{1}{8}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{5}$

Q1.22.1 MCQ 2022 1M

If $A = \begin{bmatrix} 10 & 2k + 5 \\ 3k - 3 & k + 5 \end{bmatrix}$ is a symmetric matrix, the value of k is _____.

A. 8 **B.** 5 **C.** −0.4 **D.** $\frac{1 + \sqrt{1561}}{12}$

Q1.22.2 MSQ 2022 2M

The system of linear equations in real (x, y) given by

$$\begin{pmatrix} x & y \end{pmatrix} \begin{bmatrix} 2 & 5 - 2\alpha \\ \alpha & 1 \end{bmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

involves a real parameter α and has infinitely many non-trivial solutions for special value(s) of α. Which one or more among the following options is/are non-trivial solution(s) of (x, y) for such special value(s) of α?

☐ **A.** $x = 2, \quad y = -2$

☐ **B.** $x = -1, \quad y = 4$

☐ **C.** $x = 1, \quad y = 1$

☐ **D.** $x = 4, \quad y = -2$

Q1.22.3 NAT 2022 2M

If the sum and product of eigenvalues of a 2×2 real matrix $\begin{bmatrix} 3 & p \\ p & q \end{bmatrix}$ are 4 and −1 respectively, then $|p|$ is _____ (in integer).

Q1.22.4 MSQ 2022 2M

A is a 3×5 real matrix of rank 2. For the set of homogeneous equations $A\mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is a zero vector and $\mathbf{x}$ is a vector of unknown variables, which of the following is/are true?

☐ **A.** The given set of equations will have a unique solution.

☐ **B.** The given set of equations will be satisfied by a zero vector of appropriate size.

☐ **C.** The given set of equations will have infinitely many solutions.

☐ **D.** The given set of equations will have many but a finite number of solutions.

Q1.21.1 MCQ 2021 2M ○○○

Consider a vector p in 2-dimensional space. Let its direction (counter-clockwise angle with the positive x-axis) be θ. Let p be an eigenvector of a 2×2 matrix A with corresponding eigenvalue λ, $\lambda > 0$. If we denote the magnitude of a vector v by $\|v\|$, identify the VALID statement regarding p', where $p' = Ap$.

A. Direction of $p' = \lambda\theta$, $\|p'\| = \|p\|$

B. Direction of $p' = \theta$, $\|p'\| = \lambda\|p\|$

C. Direction of $p' = \lambda\theta$, $\|p'\| = \lambda\|p\|$

D. Direction of $p' = \theta$, $\|p'\| = \|p\|/\lambda$

Q1.21.2 MCQ 2021 1M ○○○

Consider an $n \times n$ matrix A and a non-zero $n \times 1$ vector p. Their product $Ap = \alpha^2 p$, where $\alpha \in \Re$ and $\alpha \notin \{-1, 0, 1\}$. Based on the given information, the eigen value of A^2 is:

A. α **B.** α^2 **C.** $\sqrt{\alpha}$ **D.** α^4

Q1.21.3 MCQ 2021 2M ○○○

Let the superscript T represent the transpose operation. Consider the function $f(x) = \frac{1}{2}x^T Q x - r^T x$, where x and r are $n \times 1$ vectors and Q is a symmetric $n \times n$ matrix. The stationary point of $f(x)$ is

A. $Q^T r$ **B.** $Q^{-1} r$ **C.** $\dfrac{r}{r^T r}$ **D.** r

Q1.20.1 MCQ 2020 1M ○○○

Multiplication of real valued square matrices of same dimension is

A. associative

B. commutative

C. always positive definite

D. not always possible to compute

Q1.20.2 MCQ 2020 1M ○○○

A matrix P is decomposed into its symmetric part S and skew symmetric part V. If

$$S = \begin{pmatrix} -4 & 4 & 2 \\ 4 & 3 & 7/2 \\ 2 & 7/2 & 2 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & 7/2 \\ -3 & -7/2 & 0 \end{pmatrix},$$

then matrix P is

A. $\begin{pmatrix} -4 & 6 & -1 \\ 2 & 3 & 0 \\ 5 & 7 & 2 \end{pmatrix}$ **B.** $\begin{pmatrix} -4 & 2 & 5 \\ 6 & 3 & 7 \\ -1 & 0 & 2 \end{pmatrix}$

C. $\begin{pmatrix} 4 & -6 & 1 \\ -2 & -3 & 0 \\ -5 & -7 & -2 \end{pmatrix}$ **D.** $\begin{pmatrix} -2 & 9/2 & -1 \\ -1 & 81/4 & 11 \\ -2 & 45/2 & 73/4 \end{pmatrix}$

Q1.20.3 NAT 2020 1M ○○○

Let I be a 100 dimensional identity matrix and E be the set of its distinct (no value appears more than once in E) real eigenvalues. The number of elements in E is _____.

Q1.19.1 MCQ 2019 2M ○○○

The set of equations

$$x + y + z = 1$$

$$ax - ay + 3z = 5$$

$$5x - 3y + az = 6$$

has infinite solutions, if $a =$

A. -3 **B.** 3 **C.** 4 **D.** -4

Q1.19.2 MCQ 2019 1M ○○○

In matrix equation $[A]\{X\} = \{R\}$,

$$[A] = \begin{bmatrix} 4 & 8 & 4 \\ 8 & 16 & -4 \\ 4 & -4 & 15 \end{bmatrix}, \quad \{X\} = \begin{Bmatrix} 2 \\ 1 \\ 4 \end{Bmatrix}, \quad \{R\} = \begin{Bmatrix} 32 \\ 16 \\ 64 \end{Bmatrix}.$$

One of the eigenvalues of matrix $[A]$ is

A. 4 **B.** 8 **C.** 15 **D.** 16

Q1.18.1 NAT 2018 1M

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 1 \end{bmatrix}$ then $\det(A^{-1})$ is _____.
(correct to two decimal places).

Q1.17.1 MCQ 2017 2M

Consider the matrix $P = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$.
Which one of the following statements about P is INCORRECT?

A. Determinant of P is equal to 1.

B. P is orthogonal.

C. Inverse of P is equal to its transpose.

D. All Eigenvalues of P are real numbers

Q1.17.2 NAT 2017 1M

The determinant of a 2×2 matrix is 50. If one eigenvalue of the matrix is 10, the other eigenvalue is _____

Q1.17.3 NAT 2017 2M

Consider the matrix $A = \begin{bmatrix} 50 & 70 \\ 70 & 80 \end{bmatrix}$ whose eigenvectors corresponding to eigenvalues λ_1 and λ_2 are $x_1 = \begin{bmatrix} 70 \\ \lambda_1 - 50 \end{bmatrix}$ and $x_2 = \begin{bmatrix} \lambda_2 - 80 \\ 70 \end{bmatrix}$, respectively. The value of $x_1^T x_2$ is _____.

Q1.16.1 MCQ 2016 1M

The solution to the system of equations

$$\begin{bmatrix} 2 & 5 \\ -4 & 3 \end{bmatrix} \begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} 2 \\ -30 \end{Bmatrix}$$

is

A. 6, 2 **B.** −6, 2 **C.** −6, −2 **D.** 6, −2

Q1.16.2 MCQ 2016 1M

The condition for which the eigenvalues of the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & k \end{bmatrix}$$

are positive, is

A. $k > 1/2$ **B.** $k > -2$ **C.** $k > 0$ **D.** $k < -1/2$

Q1.16.3 MCQ 2016 1M

A real square matrix A is called skew-symmetric if

A. $A^T = A$ **B.** $A^T = A^{-1}$

C. $A^T = -A$ **D.** $A^T = A + A^{-1}$

Q1.16.4 NAT 2016 2M

The number of linearly independent eigenvectors of matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is _____

Q1.15.1 MCQ 2015 1M

At least one eigen value of a singular matrix is

A. Positive **B.** Zero **C.** Negative **D.** Imaginary

Q1.15.2 MCQ 2015 1M

If any two columns of a determinant $P = \begin{vmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{vmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

A. Absolute value remains unchanged but sign will change.

B. Both absolute value and sign will change.

C. Absolute value will change but sign will not change.

D. Both absolute value and sign will remain unchanged.

Q1.15.3 MCQ 2015 2M

For a given matrix $P = \begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$, where $i = \sqrt{-1}$, the inverse of matrix P is

A. $\frac{1}{24}\begin{bmatrix} 4-3i & i \\ -i & 4+3i \end{bmatrix}$ **B.** $\frac{1}{25}\begin{bmatrix} i & 4-3i \\ 4+3i & -i \end{bmatrix}$

C. $\frac{1}{24}\begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$ **D.** $\frac{1}{25}\begin{bmatrix} 4+3i & -i \\ i & 4-3i \end{bmatrix}$

Q1.14.1 MCQ 2014 1M

Given that the determinant of the matrix $\begin{bmatrix} 1 & 3 & 0 \\ 2 & 6 & 4 \\ -1 & 0 & 2 \end{bmatrix}$ is -12, the determinant of the matrix $\begin{bmatrix} 2 & 6 & 0 \\ 4 & 12 & 8 \\ -2 & 0 & 4 \end{bmatrix}$ is

A. -96 **B.** -24 **C.** 24 **D.** 96

Q1.14.2 MCQ 2014 1M

Which one of the following describes the relationship among the three vectors, $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j} + \hat{k}$ and $5\hat{i} + 6\hat{j} + 4\hat{k}$?

A. The vectors are mutually perpendicular

B. The vectors are linearly dependent

C. The vectors are linearly independent

D. The vectors are unit vectors

Q1.14.3 MCQ 2014 1M

One of the eigenvectors of the matrix $\begin{bmatrix} -5 & 2 \\ -9 & 6 \end{bmatrix}$ is

A. $\begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$ **B.** $\begin{Bmatrix} -2 \\ 9 \end{Bmatrix}$ **C.** $\begin{Bmatrix} 2 \\ -1 \end{Bmatrix}$ **D.** $\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

Q1.14.4 MCQ 2014 1M

Consider a 3×3 real symmetric matrix S such that two of its eigen values are $a \neq 0$, $b \neq 0$ with respective eigenvectors $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$. If $a \neq b$ then $x_1y_1 + x_2y_2 + x_3y_3$ equals

A. a **B.** b **C.** ab **D.** 0

Q1.14.5 MCQ 2014 1M

Which one of the following equations is a correct identity for arbitrary 3×3 real matrices P, Q and R?

A. $P(Q + R) = PQ + RP$

B. $(P - Q)^2 = P^2 - 2PQ + Q^2$

C. $\det(P + Q) = \det P + \det Q$

D. $(P + Q)^2 = P^2 + PQ + QP + Q^2$

Q1.13.1 MCQ 2013 1M

The eigen values of symmetric matrix are all

A. Complex with non-zero positive imaginary part

B. Complex with non-zero negative imaginary part

C. Real

D. Pure imaginary

Q1.13.2 MCQ 2013 1M

Choose the CORRECT set of functions, which are linearly dependent.

A. $\sin x$, $\sin^2 x$ and $\cos^2 x$ **B.** $\cos x$, $\sin x$ and $\tan x$

C. $\cos 2x$, $\sin^2 x$ and $\cos^2 x$ **D.** $\cos 2x$, $\sin x$ and $\cos x$

Q1.12.1 MCQ 2012 2M

For the matrix $A = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$, ONE of the normalized eigen vectors is given as

A. $\begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix}$ **B.** $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$ **C.** $\begin{pmatrix} \frac{3}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} \end{pmatrix}$ **D.** $\begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$

Q1.10.1 MCQ 2010 1M

One of the eigen vectors of the matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$ is

A. $\begin{Bmatrix} 2 \\ -1 \end{Bmatrix}$ **B.** $\begin{Bmatrix} 2 \\ 1 \end{Bmatrix}$ **C.** $\begin{Bmatrix} 4 \\ 1 \end{Bmatrix}$ **D.** $\begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$

Q1.09.1 MCQ 2009 1M

For a matrix $[M] = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$, the transpose of the matrix is equal to the inverse of the matrix $[M]^T = [M]^{-1}$. The value of x is given by

A. $-\frac{4}{5}$ **B.** $-\frac{3}{5}$ **C.** $\frac{3}{5}$ **D.** $\frac{4}{5}$

Q1.08.1 MCQ 2008 1M

The matrix $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 0 & 6 \\ 1 & 1 & p \end{bmatrix}$ has one eigenvalue equal to 3.

The sum of the other two eigenvalues is

A. p **B.** $p - 1$ **C.** $p - 2$ **D.** $p - 3$

Q1.08.2 MCQ 2008 2M

The eigenvectors of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ are written in the form $\begin{bmatrix} 1 \\ a \end{bmatrix}$ and $\begin{bmatrix} 1 \\ b \end{bmatrix}$. What is $a + b$?

A. 0 **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Q1.07.1 MCQ 2007 1M

If a square matrix A is real and symmetric, then the eigen values

A. are always real

B. are always real and positive

C. are always real and non-negative

D. occur in complex conjugate pairs

Q1.07.2 MCQ 2007 2M

The number of linearly independent eigen vectors of $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is

A. 0 **B.** 1 **C.** 2 **D.** infinite

Q1.06.1 MCQ 2006 1M

Eigen values of a matrix $S = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ are 5 and 1. What are the eigen values of the matrix $S^2 = SS$?

A. 1 and 25 **B.** 6 and 4

C. 5 and 1 **D.** 2 and 10

Q1.06.2 MCQ 2006 1M

Match the items in columns I and II.

columns I	columns II
(P) Singular matrix	(1) Determinant is not defined
(Q) Non-square matrix	(2) Determinant is always one
(R) Real symmetric matrix	(3) Determinant is zero
(S) Orthogonal matrix	(4) Eigen values are always real
	(5) Eigen values are nto defined

Fig. Q1.06.2

A. P - 3 Q - 1 R - 4 S - 2 **B.** P - 2 Q - 3 R - 4 S - 1

C. P - 3 Q - 2 R - 5 S - 4 **D.** P - 3 Q - 4 R - 2 S - 1

Q1.06.3 MCQ 2006 2M

Multiplication of matrices E and F is G. Matrices E and G are $E = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

What is the matrix F?

A. $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **B.** $\begin{bmatrix} \sin\theta & \cos\theta & 0 \\ -\cos\theta & \sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} \sin\theta & -\cos\theta & 0 \\ \cos\theta & \sin\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q1.05.1 MCQ 2005 1M

A is a 3×4 real matrix and $Ax = b$ is an inconsistent system of equations. The highest possible rank of A is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q1.95.1 MCQ 1995 1M

Among the following, the pair of vectors orthogonal to each other is

A. $[3, 4, 7], [3, 4, 7]$ **B.** $[1, 0, 0], [1, 1, 0]$

C. $[1, 0, 2], [0, 5, 0]$ **D.** $[1, 1, 1], [-1, -1, -1]$

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Q1.26.1 MCQ 2026 1M

Matrix P is given as

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

The TRUE option is

A. Trace of P is equal to the sum of the Eigen values of P.

B. $P^{\mathrm{T}}P$ is an identity matrix.

C. P is a skew-symmetric matrix.

D. Absolute magnitude of each Eigen value is 1.

Q1.26.2 MCQ 2026 1M

Given:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

The above system of equations represents a

A. plane **B.** line **C.** volume **D.** point

Q1.26.3 MCQ 2026 1M

Matrix A has the eigenvalues 1, 2, and 3. The Trace of A^2 is

A. 6 **B.** 14 **C.** 20 **D.** 8

Q1.26.4 NAT 2026 1M

The eigenvalues of $[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$ are $\lambda_1 = -1.547$, $\lambda_2 = 12.330$, and $\lambda_3 = 4.711$. The absolute value of the determinant of matrix A is _____ (*rounded off to two decimal places*).

Q1.25.1 MSQ 2025 2M

Pick the CORRECT eigen value(s) of the matrix $[A]$ from the following choices.

$$[A] = \begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix}$$

A. 10 **B.** 4 **C.** -2 **D.** -10

Q1.25.2 MCQ 2025 1M

Suppose λ is an eigenvalue of matrix A and x is the corresponding eigenvector. Let x also be an eigenvector of the matrix $B = A - 2I$, where I is the identity matrix. Then, the eigenvalue of B corresponding to the eigenvector x is equal to

A. λ **B.** $\lambda + 2$ **C.** 2λ **D.** $\lambda - 2$

Q1.25.3 MCQ 2025 1M

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ -2 & -3 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. For $Ax = b$ to be solvable, which one of the following options is the correct condition on b_1, b_2, and b_3:

A. $b_1 + b_2 + b_3 = 1$

B. $3b_1 + b_2 + 2b_3 = 0$

C. $b_1 + 3b_2 + b_3 = 2$

D. $b_1 + b_2 + b_3 = 2$

Q1.24.1 MCQ 2024 1M

The statements P and Q are related to matrices A and B, which are conformable for both addition and multiplication.

P: $(A + B)^T = A^T + B^T$

Q: $(AB)^T = A^T B^T$

Which one of the following options is CORRECT?

A. P is TRUE and Q is FALSE

B. Both P and Q are TRUE

C. P is FALSE and Q is TRUE

D. Both P and Q are FALSE

Q1.24.2 NAT 2024 2M

Consider two matrices $A = \begin{bmatrix} 2 & 1 & 4 \\ 1 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 2 & 3 \\ 1 & 4 \end{bmatrix}$. The determinant of the matrix AB is _____ (in integer).

Q1.23.1 MSQ 2023 1M

If M is an arbitrary real $n \times n$ matrix, then which of the following matrices will have non-negative eigenvalues?

A. M^2 **B.** MM^T **C.** $M^T M$ **D.** $(M^T)^2$

Q1.23.2 MSQ 2023 2M

For the matrix

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

which of the following statements is/are TRUE?

☐ **A.** The eigenvalues of $[A]^T$ are same as the eigenvalues of $[A]$

☐ **B.** The eigenvalues of $[A]^{-1}$ are the reciprocals of the eigenvalues of $[A]$

☐ **C.** The eigenvectors of $[A]^T$ are same as the eigenvectors of $[A]$

☐ **D.** The eigenvectors of $[A]^{-1}$ are same as the eigenvectors of $[A]$

Q1.23.3 MCQ 2023 2M

Two vectors $\begin{bmatrix}2 & 1 & 0 & 3\end{bmatrix}^T$ and $\begin{bmatrix}1 & 0 & 1 & 2\end{bmatrix}^T$ belong to the null space of a 4×4 matrix of rank 2. Which one of the following vectors also belongs to the null space?

A. $\begin{bmatrix}1 & 1 & -1 & 1\end{bmatrix}^T$ **B.** $\begin{bmatrix}2 & 0 & 1 & 2\end{bmatrix}^T$

C. $\begin{bmatrix}0 & -2 & 1 & -1\end{bmatrix}^T$ **D.** $\begin{bmatrix}3 & 1 & 1 & 2\end{bmatrix}^T$

Q1.23.4 MCQ 2023 2M

Cholesky decomposition is carried out on the following square matrix $[A]$.

$$[A] = \begin{bmatrix}8 & -5 \\ -5 & a_{22}\end{bmatrix}$$

Let l_{ij} and a_{ij} be the $(i, j)^{\text{th}}$ elements of matrices $[L]$ and $[A]$, respectively. If the element l_{22} of the decomposed lower triangular matrix $[L]$ is 1.968, what is the value (rounded off to the nearest integer) of the element a_{22}?

A. 5 **B.** 7 **C.** 9 **D.** 11

Q1.22.1 MSQ 2022 1M

The matrix M is defined as

$$M = \begin{bmatrix}1 & 3 \\ 4 & 2\end{bmatrix}$$

and has eigenvalues 5 and -2. The matrix Q is formed as

$$Q = M^3 - 4M^2 - 2M$$

Which of the following is/are the eigenvalue(s) of matrix Q?

A. 15 **B.** 25 **C.** -20 **D.** -30

Q1.22.2 MSQ 2022 1M

P and Q are two square matrices of the same order. Which of the following statement(s) is/are correct?

☐ **A.** If P and Q are invertible, then $[PQ]^{-1} = Q^{-1}P^{-1}$.

☐ **B.** If P and Q are invertible, then $[QP]^{-1} = P^{-1}Q^{-1}$.

☐ **C.** If P and Q are invertible, then $[PQ]^{-1} = P^{-1}Q^{-1}$.

☐ **D.** If P and Q are not invertible, then $[PQ]^{-1} = Q^{-1}P^{-1}$.

Q1.22.3 NAT 2022 1M

The components of pure shear strain in a sheared material are given in the matrix form:

$$\varepsilon = \begin{bmatrix}1 & 1 \\ 1 & -1\end{bmatrix}$$

Here, $\text{Trace}(\varepsilon) = 0$. Given, $P = \text{Trace}(\varepsilon^8)$ and $Q = \text{Trace}(\varepsilon^{11})$. The numerical value of $(P + Q)$ is _____. (*in integer*)

Q1.22.4 MSQ 2022 2M

Let y be a non-zero vector of size 2022×1. Which of the following statement(s) is/are TRUE?

☐ **A.** yy^T is a symmetric matrix.

☐ **B.** y^Ty is an eigenvalue of yy^T.

☐ **C.** yy^T has a rank of 2022.

☐ **D.** yy^T is invertible.

Q1.20.1 NAT 2020 2M

Consider the system of equations

$$\begin{bmatrix}1 & 3 & 2 \\ 2 & 2 & -3 \\ 4 & 4 & -6 \\ 2 & 5 & 2\end{bmatrix}\begin{bmatrix}x_1 \\ x_2 \\ x_3\end{bmatrix} = \begin{bmatrix}1 \\ 1 \\ 2 \\ 1\end{bmatrix}$$

The value of x_3 (*round off to the nearest integer*), is _____.

Q1.20.2 MCQ 2020 1M

A 4×4 matrix $[P]$ is given below

$$[P] = \begin{bmatrix}0 & 1 & 3 & 0 \\ -2 & 3 & 0 & 4 \\ 0 & 0 & 6 & 1 \\ 0 & 0 & 1 & 6\end{bmatrix}$$

The eigenvalues of $[P]$ are

A. 0, 3, 6, 6 **B.** 1, 2, 3, 4

C. 3, 4, 5, 7 **D.** 1, 2, 5, 7

Q1.19.1 MCQ 2019 1M

Euclidean norm (length) of the vector $\begin{bmatrix}4 & -2 & -6\end{bmatrix}^T$ is

A. $\sqrt{12}$ **B.** $\sqrt{24}$ **C.** $\sqrt{48}$ **D.** $\sqrt{56}$

Q1.18.1 MCQ 2018 1M

For the given orthogonal matrix Q,

$$Q = \begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$$

The inverse is

A. $\begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$

B. $\begin{bmatrix} -3/7 & -2/7 & -6/7 \\ 6/7 & -3/7 & -2/7 \\ -2/7 & -6/7 & 3/7 \end{bmatrix}$

C. $\begin{bmatrix} 3/7 & -6/7 & 2/7 \\ 2/7 & 3/7 & 6/7 \\ 6/7 & 2/7 & -3/7 \end{bmatrix}$

D. $\begin{bmatrix} -3/7 & 6/7 & -2/7 \\ -2/7 & -3/7 & -6/7 \\ -6/7 & -2/7 & 3/7 \end{bmatrix}$

Q1.18.2 MCQ 2018 1M

The matrix $\begin{pmatrix} 2 & -4 \\ 4 & -2 \end{pmatrix}$ has

A. real eigenvalues and eigenvectors

B. real eigenvalues but complex eigenvectors

C. complex eigenvalues but real eigenvectors

D. complex eigenvalues and eigenvectors

Q1.17.1 MCQ 2017 1M

Consider the following simultaneous equations (with c_1 and c_2 being constants):

$$3x_1 + 2x_2 = c_1$$
$$4x_1 + x_2 = c_2$$

The characteristic equation for these simultaneous equations is

A. $\lambda^2 - 4\lambda - 5 = 0$ **B.** $\lambda^2 - 4\lambda + 5 = 0$

C. $\lambda^2 + 4\lambda - 5 = 0$ **D.** $\lambda^2 + 4\lambda + 5 = 0$

Q1.17.2 MCQ 2017 2M

If $A = \begin{bmatrix} 1 & 5 \\ 6 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 7 \\ 8 & 4 \end{bmatrix}$, AB^{T} is equal to

A. $\begin{bmatrix} 38 & 28 \\ 32 & 56 \end{bmatrix}$ **B.** $\begin{bmatrix} 3 & 40 \\ 42 & 8 \end{bmatrix}$

C. $\begin{bmatrix} 43 & 27 \\ 34 & 50 \end{bmatrix}$ **D.** $\begin{bmatrix} 38 & 32 \\ 28 & 56 \end{bmatrix}$

Q1.17.3 MCQ 2017 1M

The matrix P is the inverse of a matrix Q. If I denotes the identity matrix, which one of the following options is correct?

A. $PQ = I$ but $QP \neq I$

B. $QP = I$ but $PQ \neq I$

C. $PQ = I$ and $QP = I$

D. $PQ - QP = I$

Q1.17.4 MCQ 2017 2M

Consider the matrix $\begin{bmatrix} 5 & -1 \\ 4 & 1 \end{bmatrix}$. Which one of the following statements is TRUE for the eigenvalues and eigenvectors of this matrix?

A. Eigenvalue 3 has a multiplicity of 2, and only one independent eigenvector exists.

B. Eigenvalue 3 has a multiplicity of 2, and two independent eigenvectors exist.

C. Eigenvalue 3 has a multiplicity of 2, and no independent eigenvector exists.

D. Eigenvalues are 3 and −3, and two independent eigenvectors exist.

Q1.16.1 MCQ 2016 1M

If the entries in each column of a square matrix M add up to 1, then an eigenvalue of M is

A. 4 **B.** 3 **C.** 2 **D.** 1

Q1.15.1 MCQ 2015 1M

Let $A = [a_{ij}]$, $1 \le i, j \le n$ with $n \ge 3$ and $a_{ij} = i.j$. The rank of A is:

A. 0 **B.** 1 **C.** n-1 **D.** n

Q1.15.2 MCQ 2015 2M

The two Eigen values of the matrix $\begin{bmatrix} 2 & 1 \\ 1 & p \end{bmatrix}$ have a ratio of 3 : 1 for $p = 2$. What is another value of p for which the Eigen values have the same ratio of 3 : 1?

A. -2 **B.** 1 **C.** $\frac{7}{3}$ **D.** $\frac{14}{3}$

Q1.14.1 NAT 2014 1M

Given the matrices $J = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 6 \end{bmatrix}$ and $K = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, the product $K^{\mathrm{T}}JK$ is _____

Q1.13.1 NAT 2013 1M

What is the minimum number of multiplications involved in computing the matrix product PQR? Matrix P has 4 rows and 2 columns, matrix Q has 2 rows and 4 columns, and matrix R has 4 rows and 1 column._____

Q1.11.1 MCQ 2011 1M

$[A]$ is a square matrix which is neither symmetric nor skew-symmetric and $[A]^{\mathrm{T}}$ is it's transpose. The sum and difference of these matrices are defined as $[S] = [A]+[A]^{\mathrm{T}}$ and $[D] = [A]-[A]^{\mathrm{T}}$ respectively. Which of the following statements is true?

A. Both $[S]$ and $[D]$ are symmetric

B. Both $[S]$ and $[D]$ are skew-symmetric

C. $[S]$ is skew-symmetric and $[D]$ is symmetric

D. $[S]$ is symmetric and $[D]$ is skew-symmetric

Q1.10.1 MCQ 2010 2M

The inverse of the matrix $\begin{bmatrix} 3+2i & i \\ -i & 3-2i \end{bmatrix}$ is

A. $\frac{1}{12}\begin{bmatrix} 3+2i & -i \\ i & 3-2i \end{bmatrix}$ **B.** $\frac{1}{12}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

C. $\frac{1}{14}\begin{bmatrix} 3+2i & -i \\ i & 3-2i \end{bmatrix}$ **D.** $\frac{1}{14}\begin{bmatrix} 3-2i & -i \\ i & 3+2i \end{bmatrix}$

Q1.09.1 MCQ 2009 2M

In the solution of the following set of linear equations by Gauss elimination using partial pivoting $5x + y + 2z = 34$; $4y - 3z = 12$; and $10x - 2y + z = -4$. The pivots for elimination of x and y are

A. 10 and 4 **B.** 10 and 2 **C.** 5 and 4 **D.** 5 and -4

Q1.08.1 MCQ 2008 1M

The product of matrices $(PQ)^{-1}P$ is

A. P^{-1} **B.** Q^{-1} **C.** $P^{-1}Q^{-1}$ **D.** PQP^{-1}

Q1.05.1 MCQ 2005 1M

Consider the matrices $X_{(4,3)}$, $Y_{(4,3)}$ and $P_{(2,3)}$. The order of $\left(P(X^{\mathrm{T}}Y)^{-1}P^{\mathrm{T}}\right)^{\mathrm{T}}$ will be

A. (2×2) **B.** (3×3) **C.** (4×3) **D.** (3×4)

Q1.05.2 MCQ 2005 1M

Consider a non-homogeneous system of linear equations representing mathematically an over-determined system. Such a system will be

A. Consistent having a unique solution

B. Consistent having many solutions

C. Inconsistent having unique solution

D. Inconsistent having no solution

Q1.05.3 MCQ 2005 2M

Consider the system of equations $A_{(n\times n)}X_{(n\times 1)} = l_{(n\times 1)}$ where, 1 is a scalar. Let (l_i, x_i) be an eigen-pair of an eigen value and its corresponding eigen vector for real matrix A. Let I be a $(n \times n)$ unit matrix. Which one of the following statement is NOT correct?

A. For a homogeneous $n \times n$ system of linear equations $(A - l_iI)x = 0$ having a nontrivial solution, the rank of $(A - l_iI)$ is less than n.

B. For matrix A^m, m being a positive integer, (λ_i^x, x_i^m) will be the eigen-pair for all i.

C. If $A^{\mathrm{T}} = A^{-1}$, then $|l_i| = 1$ for all i.

D. If $A^{\mathrm{T}} = A$, then l_i is real for all i.

Q1.04.1 MCQ 2004 1M

Real matrices $\{A\}_{3\times1}$, $[B]_{3\times3}$, $[C]_{3\times5}$, $[D]_{5\times3}$, $[E]_{5\times5}$, and $[F]_{5\times1}$ are given. Matrices $[B]$ and $[E]$ are symmetric. Following statements are made with respect to these matrices.

(I) Matrix product $\{F\}^T[C]^T[B][C]\{F\}$ is a scalar

(II) Matrix product $[D]^T[F][D]$ is always symmetric

With reference to above statements, which of the following applies?

A. Statement I is true but II is false

B. Statement I is false but II is true

C. Both the statements are true

D. Both the statements are false

Q1.01.1 MCQ 2001 2M

The product $[P][Q]^T$ of the following two matrices $[P]$ and $[Q]$ is

$$[P] = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}; [Q] = \begin{bmatrix} 4 & 8 \\ 9 & 2 \end{bmatrix}$$

A. $\begin{bmatrix} 32 & 24 \\ 56 & 46 \end{bmatrix}$ **B.** $\begin{bmatrix} 46 & 56 \\ 24 & 32 \end{bmatrix}$

C. $\begin{bmatrix} 35 & 22 \\ 61 & 42 \end{bmatrix}$ **D.** $\begin{bmatrix} 32 & 56 \\ 24 & 46 \end{bmatrix}$

Q1.00.1 MCQ 2000 1M

Consider the following two statements :

(I) The maximum number of linearly independent column vectors of a matrix A is called the rank of A.

(II) If A is an $n \times n$ square matrix, it will be nonsingular is rank $A = n$.

With reference to the above statements, which of the following applies ?

A. Both the statements are false

B. Both the statements are true.

C. I is true but II is false.

D. I is false but II is true.

Q1.99.1 MCQ 1999 1M

If A is any $n \times n$ matrix and k is a scalar, $|kA| = \alpha|A|$, where α is

A. kn **B.** n^k **C.** k^n **D.** k/n

Q1.99.2 MCQ 1999 1M

Number of terms in the expansion of general determinant of order n is

A. n^2 **B.** $n!$ **C.** n **D.** $(n+1)^2$

Q1.99.3 MCQ 1999 2M

The equation $\begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & -1 \\ y & x^2 & x \end{vmatrix} = 0$, represents a parabola passing through the points

A. $(0,1), (0,2), (0,-1)$ **B.** $(0,0), (-1,1), (1,2)$

C. $(1,1), (0,0), (2,2)$ **D.** $(1,2), (2,1), (0,0)$

Q1.98.1 MCQ 1998 1M

In matrix algebra $AS = AT$ (A, S, T, are matrices of appropriate order) implies $S = T$ only if

A. A is symmetric **B.** A is singular

C. A is non singular **D.** A is skew symmetric

Q1.98.2 MCQ 1998 2M

The real symmetric matrix C corresponding to the Quadratic form

$$Q = 4x_1x_2 - 5x_2^2, \text{ is}$$

A. $\begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix}$ **B.** $\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & 2 \\ 2 & -5 \end{bmatrix}$

Q1.97.1 MCQ 1997 1M

If A and B are two matrices and if AB exists, then BA exists

A. only if A has as many rows as B has columns

B. only if both A and B are square matrices

C. only if A and B are skew matrices

D. only if both A and B are symmetric

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Q1.26.1 MCQ 2026 1M

Which one of the following is the pair of eigenvalues of the matrix $\begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$?

A. $-7, 7$ **B.** $-5, 5$ **C.** $-4, 3$ **D.** $-3, 4$

Q1.24.1 MSQ 2024 1M

Consider a linear homogeneous system of equations $\mathbf{Ax} = \mathbf{0}$, where $\mathbf{A}$ is an $n \times n$ matrix, $\mathbf{x}$ is an $n \times 1$ vector and $\mathbf{0}$ is an $n \times 1$ null vector. Let r be the rank of $\mathbf{A}$. For a non-trivial solution to exist, which of the following conditions is/are satisfied?

A. Determinant of $\mathbf{A} = 0$

B. $r = n$

C. $r < n$

D. Determinant of $\mathbf{A} \neq 0$

Q1.24.2 NAT 2024 1M

Consider a matrix $\mathbf{A} = \begin{bmatrix} -5 & a \\ -2 & -2 \end{bmatrix}$, where a is a constant. If the eigenvalues of $\mathbf{A}$ are -1 and -6, then the value of a, rounded off to the nearest integer, is _____

Q1.23.1 NAT 2023 2M

If a matrix M is defined as $M = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$, the sum of all the eigenvalues of M^3 is equal to _____ (in integer).

Q1.22.1 MCQ 2022 1M

Given matrix $A = \begin{bmatrix} x & 1 & 3 \\ y & 2 & 6 \\ 3 & 5 & 7 \end{bmatrix}$, the ordered pair (x, y) for which $\det(A) = 0$ is

A. $(1, 1)$ **B.** $(1, 2)$

C. $(2, 2)$ **D.** $(2, 1)$

Q1.21.1 NAT 2021 1M

$\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are vectors of length 4.

$$\mathbf{A} = [a_1 \quad a_2 \quad a_3 \quad a_4]$$
$$\mathbf{B} = [b_1 \quad b_2 \quad b_3 \quad b_4]$$
$$\mathbf{C} = [c_1 \quad c_2 \quad c_3 \quad c_4]$$
$$\mathbf{D} = [d_1 \quad d_2 \quad d_3 \quad d_4]$$

It is known that $\mathbf{B}$ is not a scalar multiple of $\mathbf{A}$. Also, $\mathbf{C}$ is linearly independent of $\mathbf{A}$ and $\mathbf{B}$. Further, $\mathbf{D} = 3\mathbf{A} + 2\mathbf{B} + \mathbf{C}$. The rank of the matrix $\begin{bmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{bmatrix}$ is _____.

Q1.21.2 MCQ 2021 2M

Let A be a square matrix of size $n \times n$ ($n > 1$). The elements of $A = \{a_{ij}\}$ are given by

$$a_{ij} = \begin{cases} i \times j, & \text{if } i \geq j \\ 0, & \text{if } i < j \end{cases}$$

The determinant of A is

A. 0 **B.** 1 **C.** $n!$ **D.** $(n!)^2$

Q1.19.1 MCQ 2019 1M

A system of n homogeneous linear equations containing n unknowns will have non-trivial solutions if and only if the determinant of the coefficient matrix is

A. 1 **B.** -1 **C.** 0 **D.** ∞

Q1.18.1 NAT 2018 2M

For the matrix $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ if det stands for the determinant and A^T is the transpose of A then the value of $\det(A^T A)$ is _____

Q1.16.1 NAT 2016 2M

A set of simultaneous linear algebraic equations is represented in a matrix form as shown below.

$$\begin{bmatrix} 0 & 0 & 0 & 4 & 13 \\ 2 & 5 & 5 & 2 & 10 \\ 0 & 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 4 & 5 \\ 2 & 3 & 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 46 \\ 161 \\ 61 \\ 30 \\ 81 \end{bmatrix}$$

The value (rounded off to the nearest integer) of x_3 is _____

Q1.15.1 MCQ 2015 1M

The following set of three vectors

$$\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}x\\6\\x\end{pmatrix} \text{ and } \begin{pmatrix}3\\4\\2\end{pmatrix},$$

is linearly dependent when x is equal to

A. 0 **B.** 1 **C.** 2 **D.** 3

Q1.15.2 NAT 2015 1M

For the matrix $\begin{pmatrix}4 & 3\\3 & 4\end{pmatrix}$, if $\begin{pmatrix}1\\1\end{pmatrix}$ is an eigenvector, the corresponding eigenvalue is _____.

Q1.13.1 MCQ 2013 1M

Which of the following statements are TRUE?

P. The eigen values of a symmetric matrix are real

Q. The value of the determinant of an orthogonal matrix can only be +1

R. The transpose of a square matrix **A** has the same eigen values as those of **A**

S. The inverse of an '$n \times n$' matrix exists if and only if the rank is less than 'n'

A. P and Q only **B.** P and R only

C. Q and R only **D.** P and S only

Q1.12.1 MCQ 2012 1M

Consider the following (2×2) matrix

$$\begin{pmatrix}4 & 0\\0 & 4\end{pmatrix}$$

Which one of the following vectors is NOT a valid eigen vector of the above matrix?

A. $\begin{pmatrix}1\\0\end{pmatrix}$ **B.** $\begin{pmatrix}-2\\1\end{pmatrix}$ **C.** $\begin{pmatrix}4\\-3\end{pmatrix}$ **D.** $\begin{pmatrix}0\\0\end{pmatrix}$

Q1.11.1 MCQ 2011 1M

Let $\lambda_1 = -1$ and $\lambda_2 = 3$ be the Eigen values and $\underline{V_1} = \begin{pmatrix}1\\0\end{pmatrix}$ and $\underline{V_2} = \begin{pmatrix}1\\1\end{pmatrix}$ be the corresponding Eigen vectors of a real 2×2 matrix R. Given that $P = (\underline{V_1}\ \underline{V_2})$,

which ONE of the following matrices represents $P^{-1}RP$?

A. $\begin{pmatrix}0 & -1\\3 & 0\end{pmatrix}$ **B.** $\begin{pmatrix}0 & 3\\-1 & 0\end{pmatrix}$ **C.** $\begin{pmatrix}3 & 0\\0 & -1\end{pmatrix}$ **D.** $\begin{pmatrix}-1 & 0\\0 & 3\end{pmatrix}$

Q1.10.1 MCQ 2010 1M

A real $n \times n$ matrix $A = [a_{ij}]$ is defined as follows:

$$a_{ij} = i, \quad \text{if } i = j$$
$$= 0, \quad \text{otherwise}$$

The summation of all n eigenvalues of A is

A. $n(n+1)/2$ **B.** $n(n-1)/2$

C. $\dfrac{n(n+1)(2n+1)}{6}$ **D.** n^2

Q1.10.2 MCQ 2010 2M

X and **Y** are non-zero square matrices of size $n \times n$. If $\mathbf{XY} = \mathbf{0}_{n \times n}$ then

A. $|\mathbf{X}| = 0$ and $|\mathbf{Y}| \neq 0$ **B.** $|\mathbf{X}| \neq 0$ and $|\mathbf{Y}| = 0$

C. $|\mathbf{X}| = 0$ and $|\mathbf{Y}| = 0$ **D.** $|\mathbf{X}| \neq 0$ and $|\mathbf{Y}| \neq 0$

Q1.09.1 MCQ 2009 1M

The system of linear equations $Ax = 0$, where A is an $n \times n$ matrix, has a non-trivial solution ONLY if

A. rank of $A > n$ **B.** rank of $A = n$

C. rank of $A < n$ **D.** A is an identity matrix

Q1.09.2 MCQ 2009 2M

The eigenvalues of matrix $A = \begin{bmatrix}1 & 2\\4 & 3\end{bmatrix}$ are 5 and -1.

Then the eigenvalues of $-2A + 3I$ (I is a 2×2 identity matrix) are

A. -7 and 5 **B.** 7 and -5

C. $-\frac{1}{7}$ and $\frac{1}{5}$ **D.** $\frac{1}{7}$ and $-\frac{1}{5}$

Q1.07.1 MCQ 2007 2M

A and B are two 3×3 matrix such that

$$A = \begin{bmatrix}-2 & 4 & 6\\1 & 2 & 1\\0 & 4 & 4\end{bmatrix}, B \neq 0$$

and $AB = 0$. Then the rank of matrix B is

A. $r = 2$ **B.** $r < 3$ **C.** $r \leq 3$ **D.** $r = 3$

Q1.05.1 MCQ 2005 2M

The matrix A is given by,

$$A = \begin{pmatrix} 1 & 4 \\ a & 2 \end{pmatrix}$$

The eigenvalues of the matrix A are real and non-negative for the condition

A. $(-1/16) \le a \le (1/16)$

B. $(-1/2) \le a \le (1/2)$

C. $(-1/2) \le a \le (1/16)$

D. $(-1/16) \le a \le (1/2)$

Q1.02.1 NAT 2002 5M

Matrix $A = \begin{bmatrix} 0.1 & 0.5 & 0 \\ 0.8 & 0 & 0.4 \\ 0.1 & 0.5 & 0.6 \end{bmatrix}$ has the property that it satisfies $Ax = x$, for some vector x.

a. Write the characteristic equation to be solved for eigenvalues of A.

b. Based on visual observation, find one of the eigenvalues of A.

c. Find the other two eigenvalues of A.

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Q1.26.1 MCQ 2026 2M

Which ONE of the following options CORRECTLY matches the matrices to their properties?

Matrix		Property	
P	$\begin{bmatrix} -30 & 17 & 5 \\ 17 & 0 & -12 \\ 5 & -12 & 4 \end{bmatrix}$	1	Singular
Q	$\begin{bmatrix} 0 & 1 & 9 \\ 7 & 0 & 2 \\ 12 & 3 & 0 \end{bmatrix}$	2	Triangular
R	$\begin{bmatrix} 2/3 & 1/3 & 2/3 \\ 0 & 2/3 & -1/3 \\ 0 & 4/3 & -2/3 \end{bmatrix}$	3	Symmetric
S	$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$	4	Trace free

A. P-3; Q-1; R-4; S-2 **B.** P-4; Q-1; R-3; S-2

C. P-3; Q-4; R-1; S-2 **D.** P-4; Q-1; R-2; S-3

Q1.25.1 MCQ 2025 1M

The eigenvalues of the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are

A. $-\sqrt{-1}$ and $\sqrt{-1}$ **B.** -1 and 1

C. $1 + \sqrt{-1}$ and $1 - \sqrt{-1}$ **D.** $-1 + \sqrt{-1}$ and $-1 - \sqrt{-1}$

Q1.24.1 MCQ 2024 2M

If $\mathbf{A} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ is a matrix such that $\mathbf{A}^2 = \mathbf{I}$, where $\mathbf{I}$ is an identity matrix, then which of the following is TRUE?

A. $1 + a^2 + bc = 0$ **B.** $1 - a^2 + bc = 0$

C. $1 - a^2 - bc = 0$ **D.** $1 + a^2 - bc = 0$

Q1.23.1 MCQ 2023 1M

Given matrices $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix}$. B is skew-symmetric matrix of A. B_{13} is

A. -3 **B.** -2 **C.** 2 **D.** 3

Q1.23.2 MCQ 2023 1M

Transformation matrix to translate a point P from $(10, 15)$ to $(15, 25)$ is

A. $\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q1.23.3 NAT 2023 2M

For the matrix $\begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$, eigenvalue corresponding to the eigenvector $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ is _____*(in integer).*

Q1.22.1 MCQ 2022 2M

If a matrix is squared, then

A. both eigenvalues and eigenvectors must change

B. both eigenvalues and eigenvectors are retained

C. eigenvalues get squared but eigenvectors are retained

D. eigenvalues are retained but eigenvectors change

Q1.22.2 NAT 2022 1M

Matrix A as a product of two other matrices is given by

$$A = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \end{bmatrix}.$$

The value of $\det(A)$ is _____. *[round off to nearest integer]*

Q1.21.1 MCQ 2021 2M

The eigenvalues of matrix $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$ are 5 and 10.

For matrix $B = A + \alpha I$, where α is a constant and I is 2×2 identity matrix, its eigenvalues are

A. $5, 10$ **B.** $5 + \alpha, 10 + \alpha$

C. $5 - \alpha, 10 - \alpha$ **D.** $5\alpha, 10\alpha$

Q1.19.1 MCQ 2019 1M

For any real, square and non-singular matrix **B** , the det B^{-1} is

A. zero **B.** $(\det B)^{-1}$ **C.** $-(\det B)$ **D.** $\det B$

Q1.18.1 NAT 2018 1M

The diagonal elements of a 3x3 matrix are -10, 5 and 0, respectively. If two of its eigenvalues are -15 each, the third eigenvalue is _____

Q1.17.1 MCQ 2017 1M

For an orthogonal matrix Q, the valid equality is

A. $Q^T = Q^{-1}$ **B.** $Q = Q^{-1}$

C. $Q^T = Q$ **D.** $\det(Q) = 0$

Q1.16.1 MCQ 2016 2M

The number of solutions of the simultaneous algebraic equations $y = 3x + 3$ and $y = 3x + 5$ is

A. zero **B.** 1 **C.** 2 **D.** infinite

Q1.11.1 MCQ 2011 1M

If matrix $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ and matrix $B = \begin{bmatrix} 4 & 6 \\ 5 & 9 \end{bmatrix}$, the transpose of product of these two matrices, i.e. $(AB)^T$ is equal to

A. $\begin{bmatrix} 28 & 19 \\ 34 & 47 \end{bmatrix}$ **B.** $\begin{bmatrix} 19 & 34 \\ 47 & 28 \end{bmatrix}$

C. $\begin{bmatrix} 48 & 33 \\ 28 & 19 \end{bmatrix}$ **D.** $\begin{bmatrix} 28 & 19 \\ 48 & 33 \end{bmatrix}$

Q1.10.1 MCQ 2010 1M

If $\{1, 0, -1\}^T$ is an eigenvector of the following matrix,

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

then the corresponding eigenvalue is

A. 1 **B.** 2 **C.** 3 **D.** 5

Q1.10.2 MCQ 2010 1M

The value of q for which the following set of linear algebraic equations

$$2x + 3y = 0$$

$$6x + qy = 0$$

can have non-trivial solution is

A. 2 **B.** 7 **C.** 9 **D.** 11

Q1.09.1 MCQ 2009 2M

The value of x_3 obtained by solving the following system of linear equations is

$$x_1 + 2x_2 - 2x_3 = 4$$

$$2x_1 + x_2 + x_3 = -2$$

$$-x_1 + x_2 - x_3 = 2$$

A. -12 **B.** -2 **C.** 0 **D.** 12

Q1.08.1 MCQ 2008 2M

Inverse of the matrix $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is

A. $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ **B.** $\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

C. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ **D.** $\begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}$

Q1.07.1 MCQ 2007 1M

The determinant $\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix}$ evaluates to

A. 0 **B.** $2b(b-1)$

C. $2(1-b)(1+2b)$ **D.** $3b(1+b)$

Q1.07.2 MCQ 2007 2M

If A is square symmetric real valued matrix of dimension $2n$, the eigenvalues of A are

A. $2n$ distinct real values

B. $2n$ real values, not necessarily distinct

C. n distinct pairs of complex conjugate numbers

D. n pairs of complex conjugate numbers, not necessarily distinct

Q1.05.1 MCQ 2005 1M

The eigen values of the matrix M given are 15, 3, and 0. $M = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, the value of the determinant of a matrix is

A. 20 **B.** 10 **C.** 0 **D.** -10

Q1.95.1 MCQ 1995 1M

Given matrix $L = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 5 \end{bmatrix}$ and $M = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$ then $L \times M$ is

A. $\begin{bmatrix} 8 & 1 \\ 13 & 2 \\ 22 & 5 \end{bmatrix}$ **B.** $\begin{bmatrix} 6 & 5 \\ 9 & 8 \\ 12 & 13 \end{bmatrix}$ **C.** $\begin{bmatrix} 1 & 8 \\ 2 & 13 \\ 5 & 22 \end{bmatrix}$ **D.** $\begin{bmatrix} 6 & 2 \\ 9 & 4 \\ 0 & 5 \end{bmatrix}$

Q1.94.1 MCQ 1994 1M

For the following matrix $\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ the number of positive characteristic roots is

A. one **B.** two

C. four **D.** cannot be found

Q1.94.2 MCQ 1994 1M

If for a matrix, rank equals both the number of rows and number of columns, then the matrix is called

A. Non-singular **B.** singular

C. transpose **D.** minor

Q1.94.3 MCQ 1994 1M

The matrix $\begin{bmatrix} 1 & -4 \\ 1 & -5 \end{bmatrix}$ is an inverse of the matrix $\begin{bmatrix} 5 & -4 \\ 1 & -1 \end{bmatrix}$

A. True

B. False

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q1.26.1 B,D	Q1.25.1 A	Q1.25.2 3	Q1.24.1 5	Q1.24.2 A,B	Q1.23.1 C
Q1.23.2 A	Q1.22.1 C	Q1.22.2 A	Q1.21.1 8.0	Q1.21.2 10.0	Q1.20.1 C
Q1.20.2 A	Q1.19.1 3	Q1.18.1 C	Q1.18.2 2	Q1.17.1 4	Q1.17.2 C
Q1.17.3 C	Q1.16.1 C	Q1.16.2 1	Q1.16.3 3	Q1.16.4 D	Q1.16.5 D
Q1.15.1 2	Q1.15.2 16.5 to 17.5	Q1.15.3 B	Q1.15.4 C	Q1.14.1 1	Q1.14.2 D
Q1.14.3 1	Q1.14.4 200	Q1.14.5 B	Q1.14.6 49	Q1.14.7 B	Q1.13.1 A
Q1.13.2 B	Q1.12.1 B	Q1.11.1 B	Q1.10.1 C	Q1.09.1 D	Q1.08.1 C
Q1.08.2 B	Q1.06.1 C	Q1.06.2 A	Q1.06.3 C	Q1.05.1 A	Q1.05.2 C
Q1.00.1 B	Q1.98.1 D	Q1.94.1 A	Q1.94.2 A		
ELECTRICAL ENGINEERING					
Q1.26.1 A,B,D	Q1.26.2 0	Q1.26.3 A	Q1.26.4 A,B,D	Q1.26.5 A,B,C	Q1.25.1 B
Q1.25.2 B	Q1.25.3 A	Q1.24.1 C	Q1.24.2 29	Q1.23.1 A	Q1.22.1 B
Q1.22.2 C	Q1.22.3 A	Q1.21.1 B	Q1.21.2 1	Q1.20.1 *	Q1.19.1 D
Q1.19.2 C	Q1.18.1 5.5	Q1.18.2 0.9 to 1.1	Q1.17.1 C	Q1.17.2 A	Q1.16.1 3
Q1.16.2 A	Q1.16.3 B	Q1.16.4 D	Q1.16.5 D	Q1.15.1 9	Q1.15.2 B
Q1.15.3 A	Q1.14.1 0.33	Q1.14.2 C	Q1.14.3 A	Q1.13.1 D	Q1.11.1 B
Q1.11.2 D	Q1.10.1 B	Q1.09.1 C	Q1.08.1 D	Q1.08.2 A	Q1.08.3 B
Q1.08.4 D	Q1.07.1 B	Q1.07.2 B	Q1.07.3 D	Q1.07.4 C	Q1.07.5 A
Q1.05.1 A	Q1.05.2 B	Q1.02.1 C	Q1.98.1 C	Q1.97.1 *	Q1.95.1 *

Q1.94.1 C					
INSTRUMENTATION ENGINEERING					
Q1.26.1 C	Q1.26.2 A	Q1.25.1 A	Q1.25.2 2	Q1.24.1 C	Q1.24.2 -2
Q1.23.1 A	Q1.23.2 -8	Q1.22.1 A,D	Q1.22.2 A,C	Q1.21.1 A	Q1.21.2 4
Q1.20.1 B	Q1.20.2 D	Q1.19.1 10	Q1.18.1 A	Q1.18.2 C	Q1.17.1 1
Q1.17.2 D	Q1.17.3 C	Q1.16.1 -6	Q1.15.1 C	Q1.14.1 C	Q1.13.1 B
Q1.13.2 *	Q1.11.1 B	Q1.10.1 A	Q1.10.2 C	Q1.09.1 B	Q1.09.2 C
Q1.09.3 D	Q1.07.1 B	Q1.07.2 B	Q1.06.1 C	Q1.05.1 B	Q1.01.1 B
COMPUTER SCIENCE & IT					
Q1.26.1 A	Q1.26.2 A	Q1.26.3 B,D	Q1.26.4 24	Q1.26.5 48	Q1.25.1 A,D
Q1.25.2 D	Q1.25.3 C	Q1.25.4 A	Q1.25.5 A,B,C	Q1.24.1 B	Q1.24.2 A
Q1.24.3 A,B	Q1.23.1 B	Q1.22.1 D	Q1.22.2 A,C,D	Q1.21.1 3	Q1.21.2 4
Q1.20.1 C	Q1.19.1 12	Q1.18.1 3	Q1.18.2 D	Q1.17.1 C	Q1.17.2 B
Q1.17.3 2	Q1.17.4 5	Q1.16.1 15	Q1.16.2 C	Q1.16.3 0.124 to 0.126	Q1.16.4 1500
Q1.15.1 D	Q1.15.2 B	Q1.15.3 6	Q1.15.4 0	Q1.15.5 C	Q1.14.1 0
Q1.14.2 6	Q1.14.3 2	Q1.14.4 A	Q1.13.1 A	Q1.12.1 D	Q1.11.1 C
Q1.10.1 D	Q1.08.1 *	Q1.08.2 A	Q1.07.1 C	Q1.04.1 C	Q1.04.2 B
Q1.01.1 A	Q1.00.1 A	Q1.98.1 B	Q1.97.1 C	Q1.96.1 *	Q1.96.2 A
Q1.96.3 C	Q1.95.1 A	Q1.94.1 D			
MECHANICAL ENGINEERING					
Q1.25.1 C	Q1.24.1 D	Q1.24.2 A	Q1.22.1 A	Q1.22.2 A,B	Q1.22.3 2
Q1.22.4 B,C	Q1.21.1 B	Q1.21.2 D	Q1.21.3 B	Q1.20.1 A	Q1.20.2 B

Q1.20.3 1	Q1.19.1 C	Q1.19.2 D	Q1.18.1 0.25	Q1.17.1 D	Q1.17.2 5
Q1.17.3 0	Q1.16.1 D	Q1.16.2 A	Q1.16.3 C	Q1.16.4 2	Q1.15.1 B
Q1.15.2 A	Q1.15.3 A	Q1.14.1 A	Q1.14.2 B	Q1.14.3 D	Q1.14.4 D
Q1.14.5 D	Q1.13.1 C	Q1.13.2 C	Q1.12.1 B	Q1.10.1 A	Q1.09.1 A
Q1.08.1 C	Q1.08.2 B	Q1.07.1 A	Q1.07.2 B	Q1.06.1 A	Q1.06.2 A
Q1.06.3 C	Q1.05.1 B	Q1.95.1 C			
CIVIL ENGINEERING					
Q1.26.1 A	Q1.26.2 B	Q1.26.3 B	Q1.26.4 89.00 to 91.00	Q1.25.1 A,C	Q1.25.2 D
Q1.25.3 B	Q1.24.1 A	Q1.24.2 10	Q1.23.1 B,C	Q1.23.2 A,B,D	Q1.23.3 A
Q1.23.4 B	Q1.22.1 A,C	Q1.22.2 A,B	Q1.22.3 32	Q1.22.4 A,B	Q1.20.1 3
Q1.20.2 D	Q1.19.1 D	Q1.18.1 C	Q1.18.2 D	Q1.17.1 A	Q1.17.2 A
Q1.17.3 C	Q1.17.4 A	Q1.16.1 D	Q1.15.1 B	Q1.15.2 D	Q1.14.1 23
Q1.13.1 16	Q1.11.1 D	Q1.10.1 B	Q1.09.1 A	Q1.08.1 B	Q1.05.1 A
Q1.05.2 D	Q1.05.3 B	Q1.04.1 A	Q1.01.1 A	Q1.00.1 B	Q1.99.1 C
Q1.99.2 B	Q1.99.3 B	Q1.98.1 C	Q1.98.2 D	Q1.97.1 A	
CHEMICAL ENGINEERING					
Q1.26.1 B	Q1.24.1 A,C	Q1.24.2 -2	Q1.23.1 4160	Q1.22.1 B	Q1.21.1 3
Q1.21.2 D	Q1.19.1 C	Q1.18.1 1	Q1.16.1 15	Q1.15.1 D	Q1.15.2 7
Q1.13.1 B	Q1.12.1 D	Q1.11.1 D	Q1.10.1 A	Q1.10.2 C	Q1.09.1 C
Q1.09.2 A	Q1.07.1 B	Q1.05.1 D	Q1.02.1 *		
PRODUCTION & INDUSTRIAL ENGINEERING					
Q1.26.1 C	Q1.25.1 A	Q1.24.1 C	Q1.23.1 *	Q1.23.2 A	Q1.23.3 1

Q1.22.1 C	Q1.22.2 0	Q1.21.1 B	Q1.19.1 B	Q1.18.1 25	Q1.17.1 A
Q1.16.1 A	Q1.11.1 D	Q1.10.1 A	Q1.10.2 C	Q1.09.1 B	Q1.08.1 A
Q1.07.1 A	Q1.07.2 B	Q1.05.1 C	Q1.95.1 B	Q1.94.1 B	Q1.94.2 A
Q1.94.3 A					

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